

# Bitland Confidential

## Intel Apollo Lake Platform

M/B Schematics Document

V1.4

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Reference form:

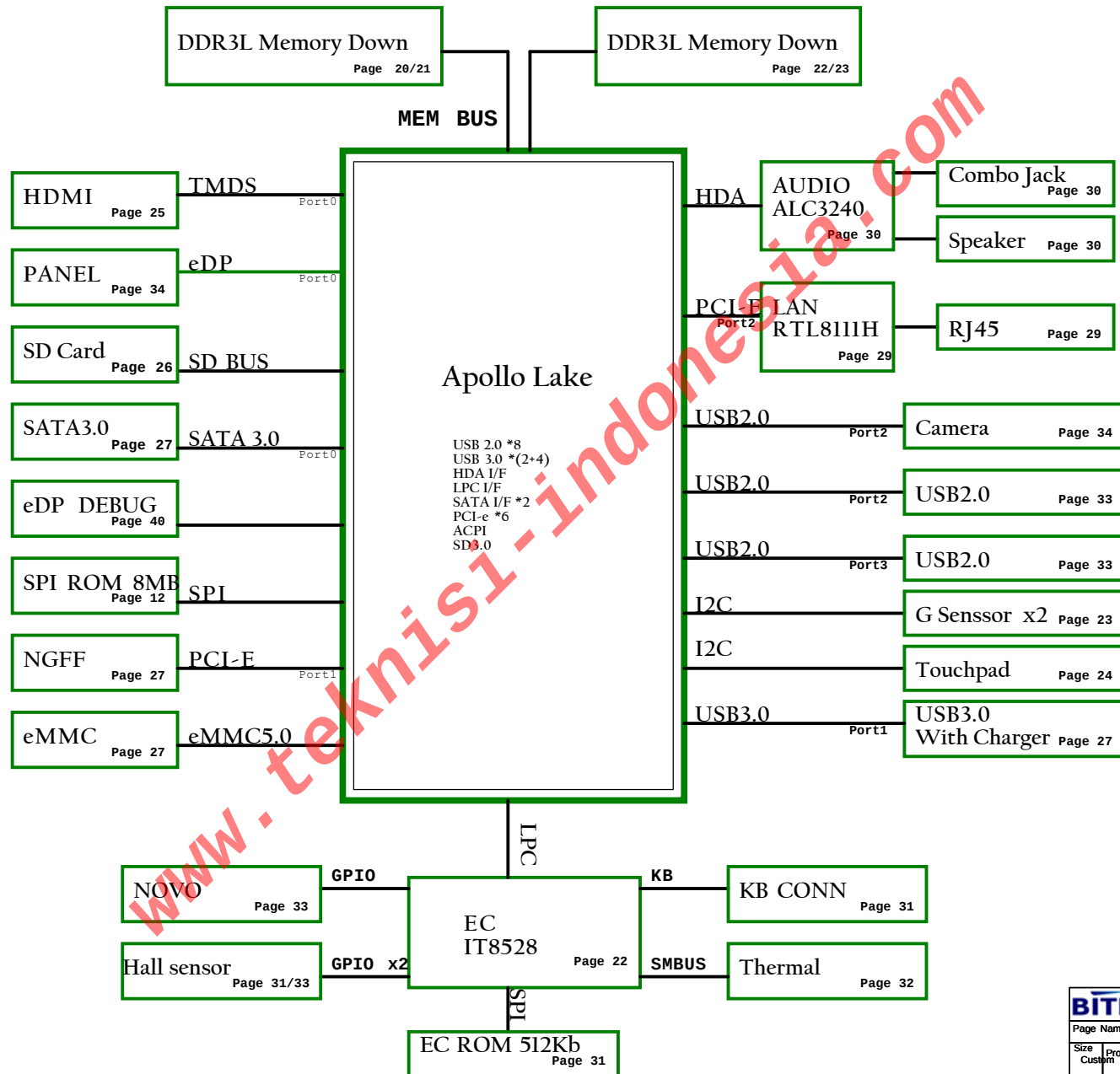
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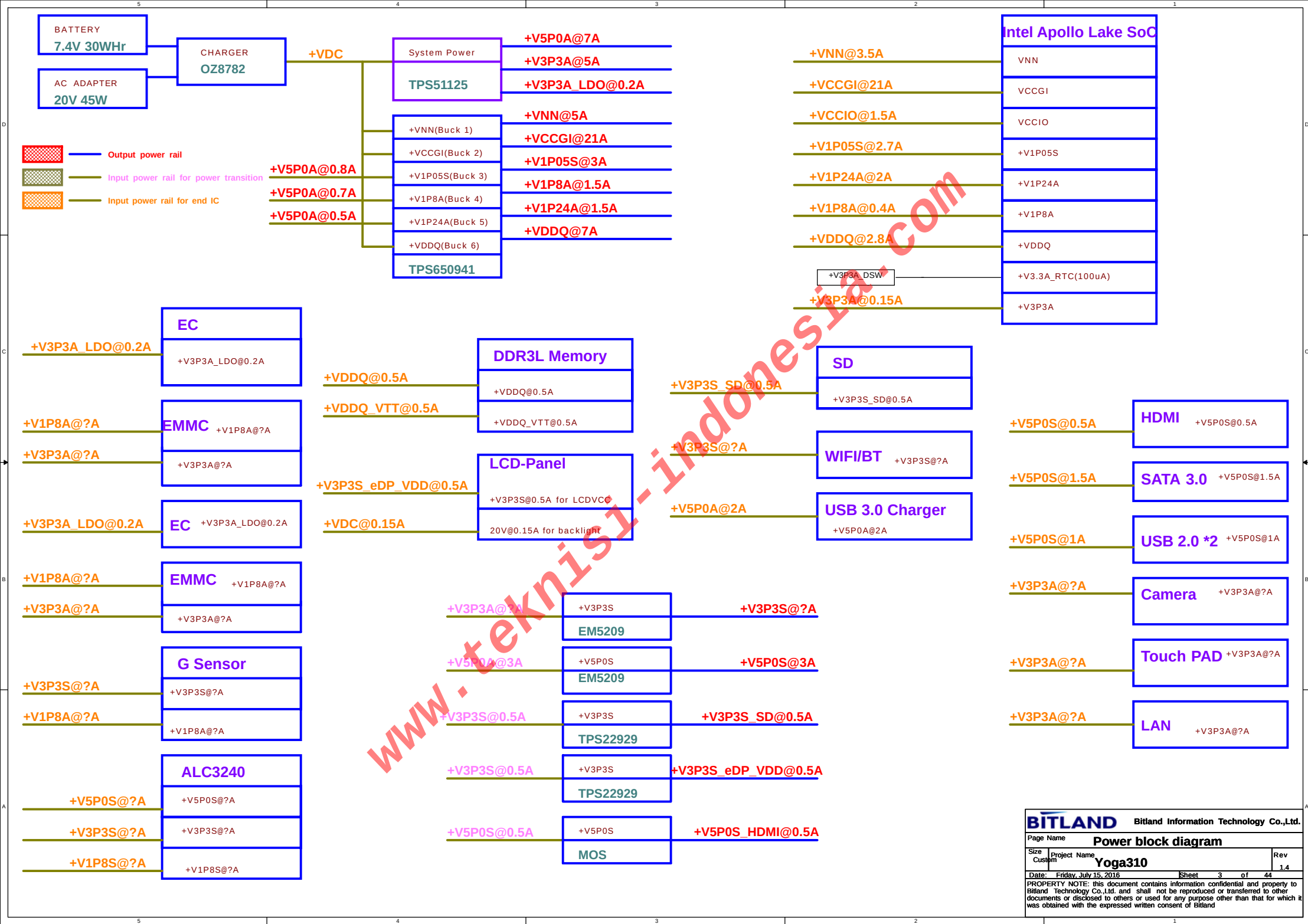
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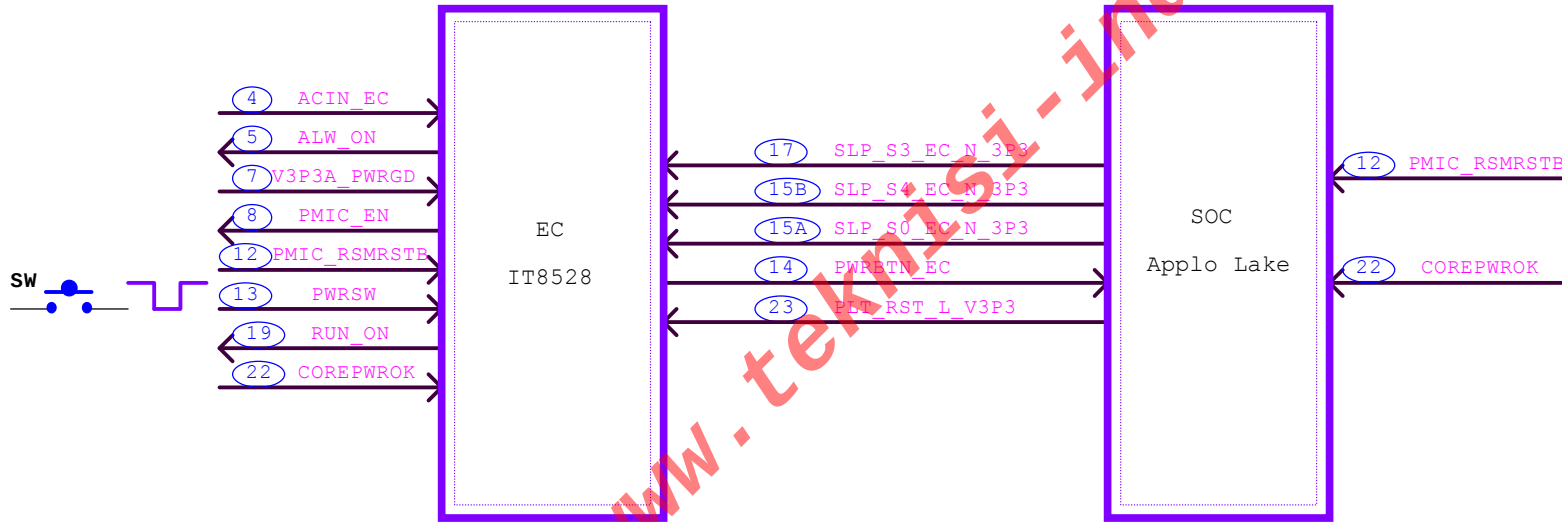
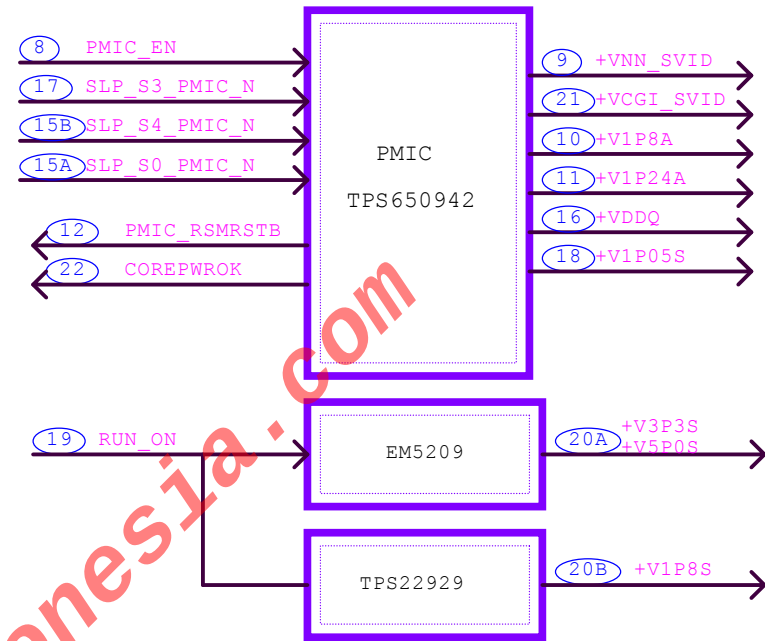
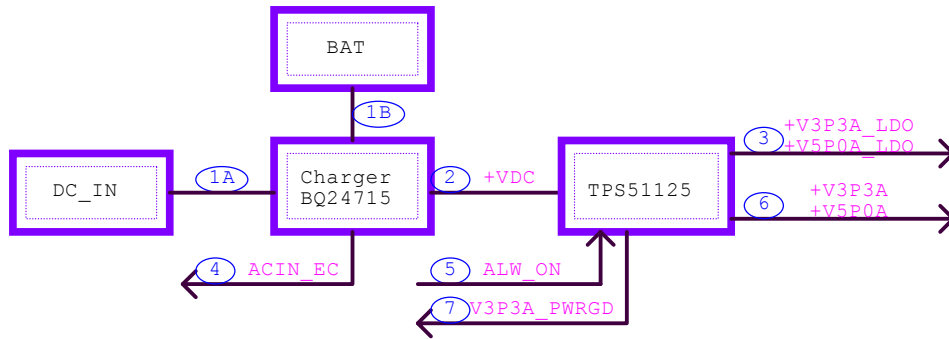
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|                                                                                                                                                                                                                                                                                                                                |                |            |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|-------|
| <b>BITLAND</b> Bitland Information Technology Co.,Ltd.                                                                                                                                                                                                                                                                         |                |            |       |
| Page Name <b>COVER PAGE</b>                                                                                                                                                                                                                                                                                                    |                |            |       |
| Size                                                                                                                                                                                                                                                                                                                           | Project Name   | Rev        |       |
| Custom                                                                                                                                                                                                                                                                                                                         | <b>Yoga310</b> | <b>1.4</b> |       |
| Date: Friday, July 15, 2016                                                                                                                                                                                                                                                                                                    | Sheet          | 1          | of 44 |
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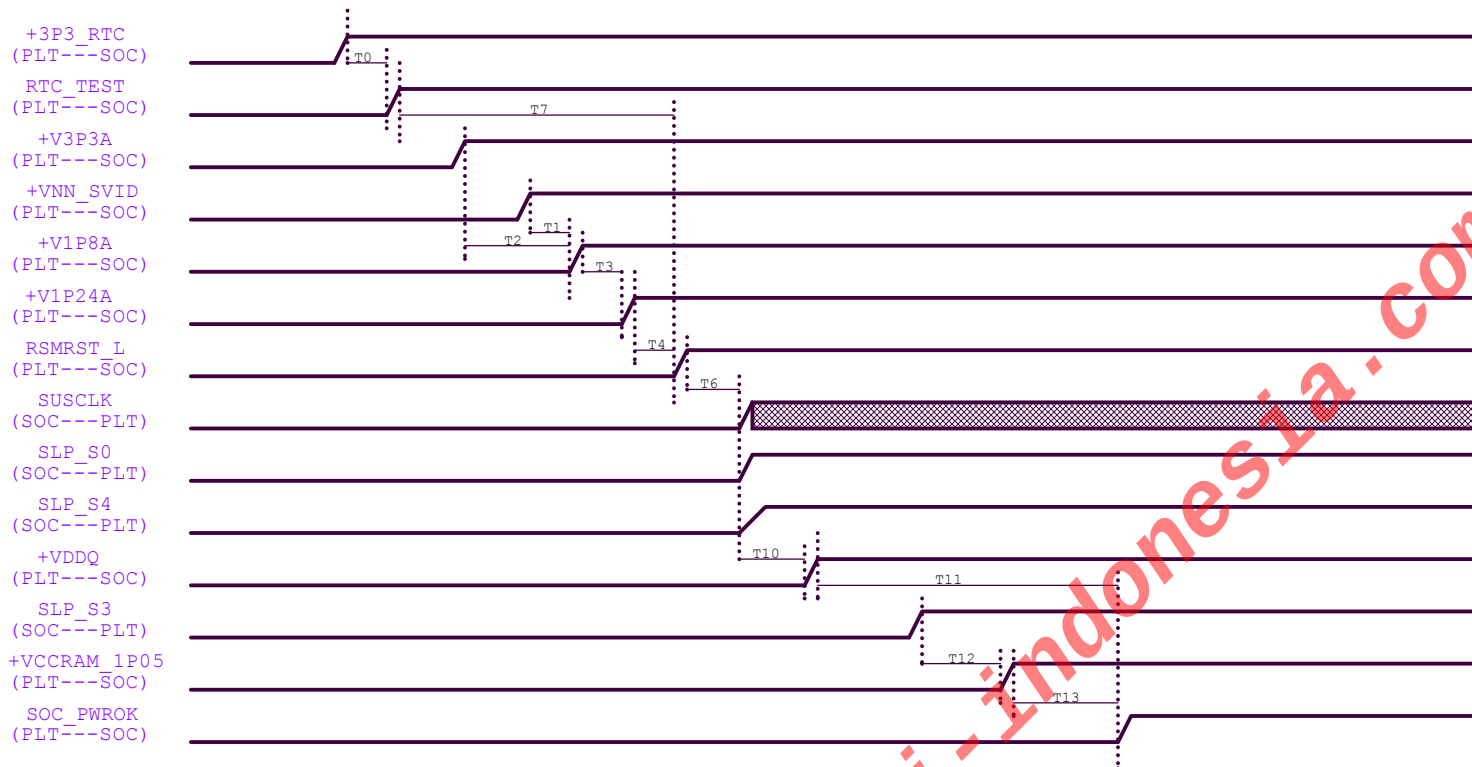
# System Block Diagram





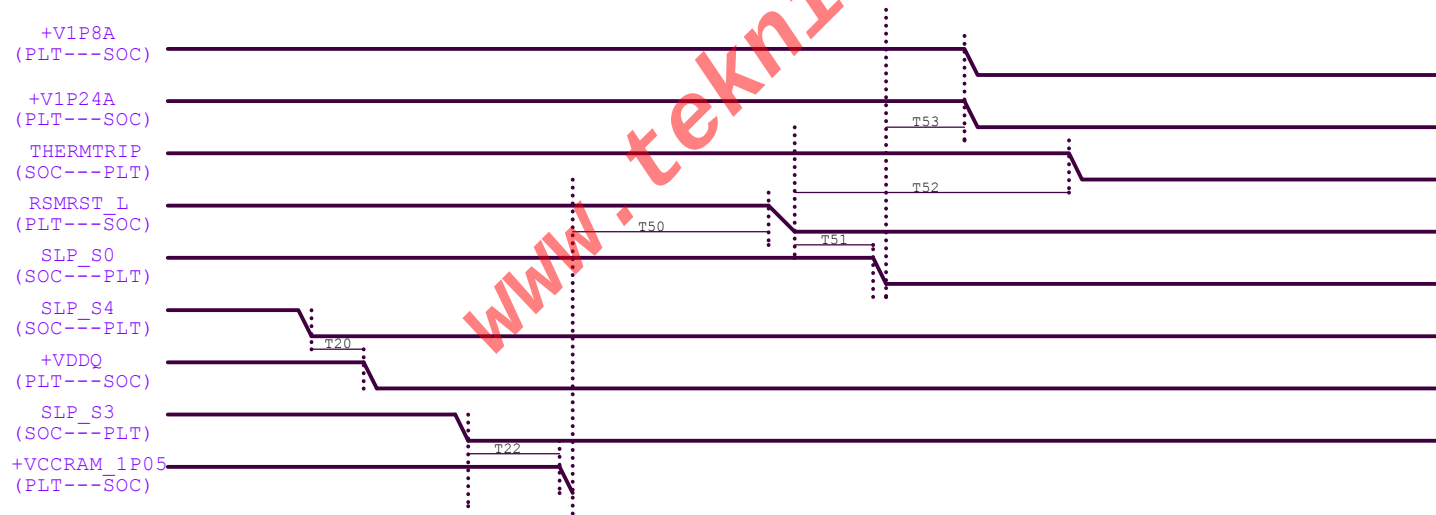


# G3-----S0



| Timing  | Minimum (ms)   |
|---------|----------------|
| T0      | 9              |
| T1      | 0              |
| T2      | 0              |
| T3      | 0              |
| T4      | 10             |
| T6      | 95             |
| T7      | 1us            |
| T10     | no requirement |
| T11     | 5              |
| T12     | no requirement |
| T13     | 5              |
| T20     | no requirement |
| T21     | no requirement |
| T22     | no requirement |
| T50     | no requirement |
| T51     | 0              |
| T52     | 2048 RTC clock |
| T51+T53 | 1us            |

# S0-----S5



| CPU1A<br>APL_SOC |              | U3E1         |               |
|------------------|--------------|--------------|---------------|
| M_A_DQ0 AY62     | MEM_CH0_DQ0  | MEM_CH0_DQ32 | AR39 M_A_DQ32 |
| M_A_DQ1 AY61     | MEM_CH0_DQ1  | MEM_CH0_DQ33 | AV37 M_A_DQ33 |
| M_A_DQ2 BE62     | MEM_CH0_DQ2  | MEM_CH0_DQ34 | AW37 M_A_DQ34 |
| M_A_DQ3 BG62     | MEM_CH0_DQ3  | MEM_CH0_DQ35 | AR37 M_A_DQ35 |
| M_A_DQ4 BD63     | MEM_CH0_DQ4  | MEM_CH0_DQ36 | AT37 M_A_DQ36 |
| M_A_DQ5 AW62     | MEM_CH0_DQ5  | MEM_CH0_DQ37 | AT41 M_A_DQ37 |
| M_A_DQ6 AW63     | MEM_CH0_DQ6  | MEM_CH0_DQ38 | AR41 M_A_DQ38 |
| M_A_DQ7 BD62     | MEM_CH0_DQ7  | MEM_CH0_DQ39 | AW35 M_A_DQ39 |
| M_A_DQ8 AV59     | MEM_CH0_DQ8  | MEM_CH0_DQ40 | BJ44 M_A_DQ40 |
| M_A_DQ9 AU63     | MEM_CH0_DQ9  | MEM_CH0_DQ41 | BG39 M_A_DQ41 |
| M_A_DQ10AU62     | MEM_CH0_DQ10 | MEM_CH0_DQ42 | BG40 M_A_DQ42 |
| M_A_DQ11AV58     | MEM_CH0_DQ11 | MEM_CH0_DQ43 | BJ40 M_A_DQ43 |
| M_A_DQ12AV57     | MEM_CH0_DQ12 | MEM_CH0_DQ44 | BG43 M_A_DQ44 |
| M_A_DQ13AT55     | MEM_CH0_DQ13 | MEM_CH0_DQ45 | BG44 M_A_DQ45 |
| M_A_DQ14AT54     | MEM_CH0_DQ14 | MEM_CH0_DQ46 | BH45 M_A_DQ46 |
| M_A_DQ15AY59     | MEM_CH0_DQ15 | MEM_CH0_DQ47 | BH41 M_A_DQ47 |
| M_A_DQ16AY57     | MEM_CH0_DQ16 | MEM_CH0_DQ48 | BA34 M_A_DQ48 |
| M_A_DQ17BB57     | MEM_CH0_DQ17 | MEM_CH0_DQ49 | BE34 M_A_DQ49 |
| M_A_DQ18BD59     | MEM_CH0_DQ18 | MEM_CH0_DQ50 | BD34 M_A_DQ50 |
| M_A_DQ19BF59     | MEM_CH0_DQ19 | MEM_CH0_DQ51 | BD37 M_A_DQ51 |
| M_A_DQ20AV54     | MEM_CH0_DQ20 | MEM_CH0_DQ52 | BB37 M_A_DQ52 |
| M_A_DQ21AY55     | MEM_CH0_DQ21 | MEM_CH0_DQ53 | BE39 M_A_DQ53 |
| M_A_DQ22AV52     | MEM_CH0_DQ22 | MEM_CH0_DQ54 | BD39 M_A_DQ54 |
| M_A_DQ23BD58     | MEM_CH0_DQ23 | MEM_CH0_DQ55 | BB34 M_A_DQ55 |
| M_A_DQ24BE56     | MEM_CH0_DQ24 | MEM_CH0_DQ56 | BJ38 M_A_DQ56 |
| M_A_DQ25BD54     | MEM_CH0_DQ25 | MEM_CH0_DQ57 | BG34 M_A_DQ57 |
| M_A_DQ26BF58     | MEM_CH0_DQ26 | MEM_CH0_DQ58 | BG33 M_A_DQ58 |
| M_A_DQ27BE50     | MEM_CH0_DQ27 | MEM_CH0_DQ59 | BH33 M_A_DQ59 |
| M_A_DQ29BB50     | MEM_CH0_DQ29 | MEM_CH0_DQ60 | BG38 M_A_DQ60 |
| M_A_DQ28BD50     | MEM_CH0_DQ28 | MEM_CH0_DQ61 | BH37 M_A_DQ61 |
| M_A_DQ30BA50     | MEM_CH0_DQ30 | MEM_CH0_DQ62 | BG37 M_A_DQ62 |
| M_A_DQ31BB54     | MEM_CH0_DQ31 | MEM_CH0_DQ63 | BJ34 M_A_DQ63 |

BROXTON\_P\_SOC\_DDR3L\_SYM  
REV = 0.7

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|                                                                                                                                                                                                                                                                                                                 |                       |                    |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|---------|
| <b>BITLAND</b> Bitland Information Technology Co.,Ltd.                                                                                                                                                                                                                                                          |                       |                    |         |
| Page Name                                                                                                                                                                                                                                                                                                       |                       | <b>SOC MEM CH0</b> |         |
| Size                                                                                                                                                                                                                                                                                                            | Project Name          | Rev                |         |
| Custom                                                                                                                                                                                                                                                                                                          | <b>Yoga310</b>        | <b>1.4</b>         |         |
| Date:                                                                                                                                                                                                                                                                                                           | Friday, July 15, 2016 | Sheet              | 6 of 44 |
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| CPU1C<br>APL_SGC |      | U3E1         |                            |
|------------------|------|--------------|----------------------------|
| M_B_DQ0          | BJ26 | MEM_CH1_DQ0  | MEM_CH1_DQ32 BF6 M_B_DQ32  |
| M_B_DQ1          | BG30 | MEM_CH1_DQ1  | MEM_CH1_DQ33 BD10 M_B_DQ33 |
| M_B_DQ2          | BH31 | MEM_CH1_DQ2  | MEM_CH1_DQ34 BE14 M_B_DQ34 |
| M_B_DQ3          | BG31 | MEM_CH1_DQ3  | MEM_CH1_DQ35 BB10 M_B_DQ35 |
| M_B_DQ4          | BH27 | MEM_CH1_DQ4  | MEM_CH1_DQ36 BA14 M_B_DQ36 |
| M_B_DQ5          | BG27 | MEM_CH1_DQ5  | MEM_CH1_DQ37 BB14 M_B_DQ37 |
| M_B_DQ6          | BG26 | MEM_CH1_DQ6  | MEM_CH1_DQ38 BD14 M_B_DQ38 |
| M_B_DQ7          | BJ30 | MEM_CH1_DQ7  | MEM_CH1_DQ39 BE8 M_B_DQ39  |
| M_B_DQ8          | BA30 | MEM_CH1_DQ8  | MEM_CH1_DQ40 AV12 M_B_DQ40 |
| M_B_DQ9          | BB30 | MEM_CH1_DQ9  | MEM_CH1_DQ41 BD6 M_B_DQ41  |
| M_B_DQ10         | BE30 | MEM_CH1_DQ10 | MEM_CH1_DQ42 BD5 M_B_DQ42  |
| M_B_DQ11         | BD30 | MEM_CH1_DQ11 | MEM_CH1_DQ43 BB7 M_B_DQ43  |
| M_B_DQ12         | BE25 | MEM_CH1_DQ12 | MEM_CH1_DQ44 AV10 M_B_DQ44 |
| M_B_DQ13         | BB27 | MEM_CH1_DQ13 | MEM_CH1_DQ45 AY9 M_B_DQ45  |
| M_B_DQ14         | BD25 | MEM_CH1_DQ14 | MEM_CH1_DQ46 AY7 M_B_DQ46  |
| M_B_DQ15         | BD27 | MEM_CH1_DQ15 | MEM_CH1_DQ47 BF5 M_B_DQ47  |
| M_B_DQ16         | BG24 | MEM_CH1_DQ16 | MEM_CH1_DQ48 AU2 M_B_DQ48  |
| M_B_DQ17         | BJ20 | MEM_CH1_DQ17 | MEM_CH1_DQ49 AT10 M_B_DQ49 |
| M_B_DQ18         | BH23 | MEM_CH1_DQ18 | MEM_CH1_DQ50 AT9 M_B_DQ50  |
| M_B_DQ19         | BJ24 | MEM_CH1_DQ19 | MEM_CH1_DQ51 AU1 M_B_DQ51  |
| M_B_DQ20         | BG20 | MEM_CH1_DQ20 | MEM_CH1_DQ52 AY5 M_B_DQ52  |
| M_B_DQ21         | BG21 | MEM_CH1_DQ21 | MEM_CH1_DQ53 AV5 M_B_DQ53  |
| M_B_DQ22         | BH19 | MEM_CH1_DQ22 | MEM_CH1_DQ54 AV6 M_B_DQ54  |
| M_B_DQ23         | BG25 | MEM_CH1_DQ23 | MEM_CH1_DQ55 AV7 M_B_DQ55  |
| M_B_DQ24         | AT27 | MEM_CH1_DQ24 | MEM_CH1_DQ56 AY2 M_B_DQ56  |
| M_B_DQ25         | AW29 | MEM_CH1_DQ25 | MEM_CH1_DQ57 BD2 M_B_DQ57  |
| M_B_DQ26         | AR27 | MEM_CH1_DQ26 | MEM_CH1_DQ58 BD1 M_B_DQ58  |
| M_B_DQ27         | AT23 | MEM_CH1_DQ27 | MEM_CH1_DQ59 BE2 M_B_DQ59  |
| M_B_DQ28         | AV27 | MEM_CH1_DQ28 | MEM_CH1_DQ60 AW1 M_B_DQ60  |
| M_B_DQ29         | AR25 | MEM_CH1_DQ29 | MEM_CH1_DQ61 AW2 M_B_DQ61  |
| M_B_DQ30         | AR23 | MEM_CH1_DQ30 | MEM_CH1_DQ62 AY3 M_B_DQ62  |
| M_B_DQ31         | AW27 | MEM_CH1_DQ31 | MEM_CH1_DQ63 BG2 M_B_DQ63  |

BROXTON\_P\_SOC\_DDR3L\_SYM  
REV = 0.7

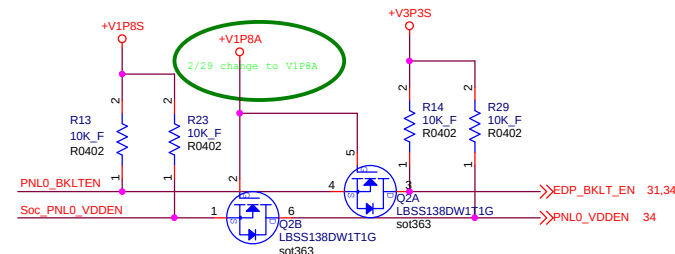
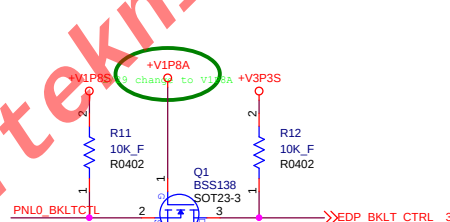
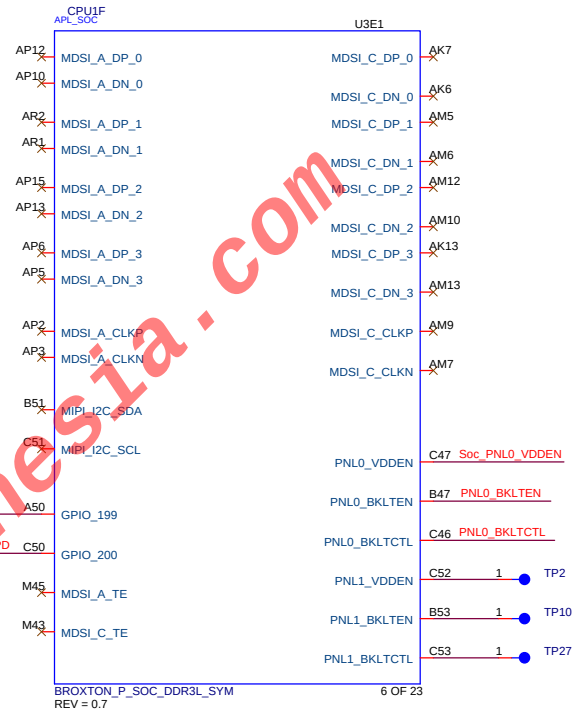
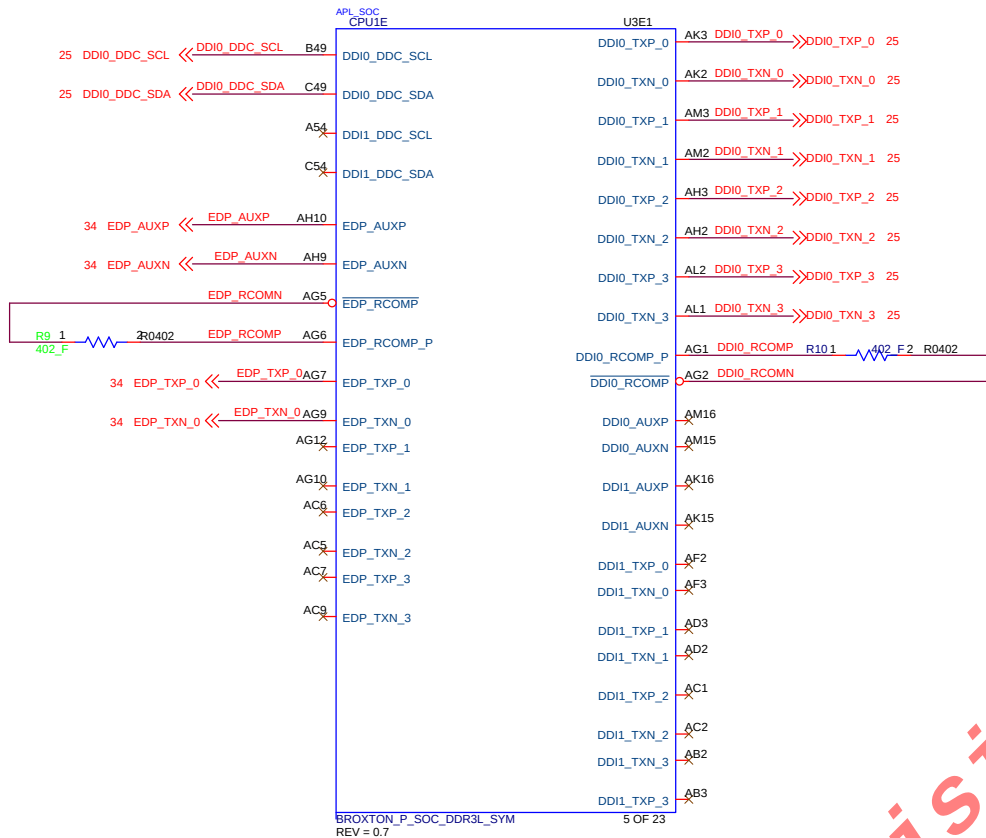
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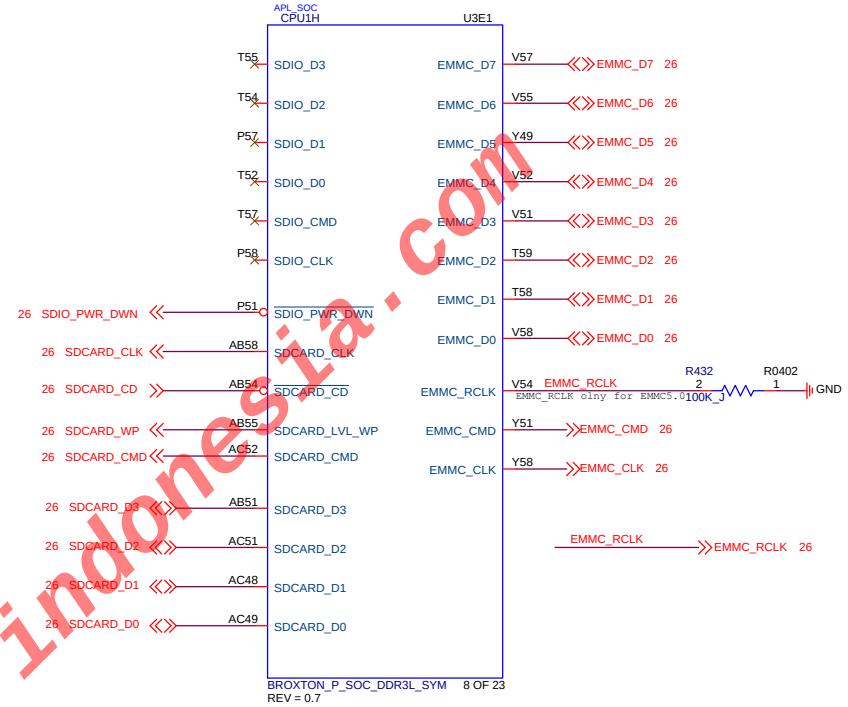
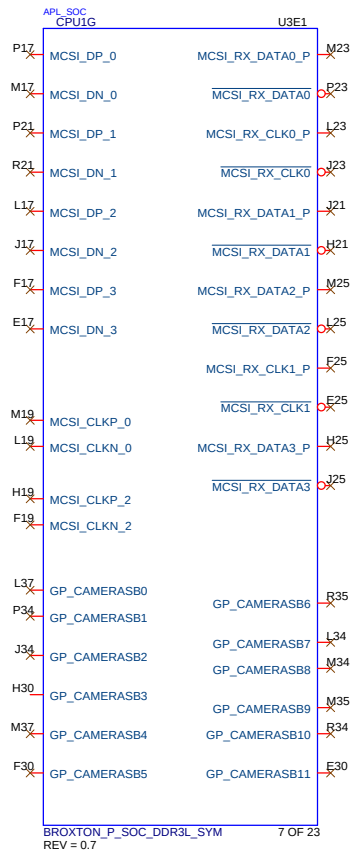
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| <b>BITLAND</b> Bitland Information Technology Co.,Ltd.                                                                                                                                                                                                                                                          |                       |       |            |
| Page Name <b>SOC MEM CH1</b>                                                                                                                                                                                                                                                                                    |                       |       |            |
| Size                                                                                                                                                                                                                                                                                                            | Project Name          |       | Rev        |
| Custom                                                                                                                                                                                                                                                                                                          | <b>Yoga310</b>        |       | <b>1.4</b> |
| Date:                                                                                                                                                                                                                                                                                                           | Friday, July 15, 2016 | Sheet | 8 of 44    |
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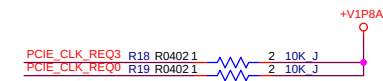
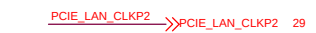
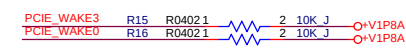
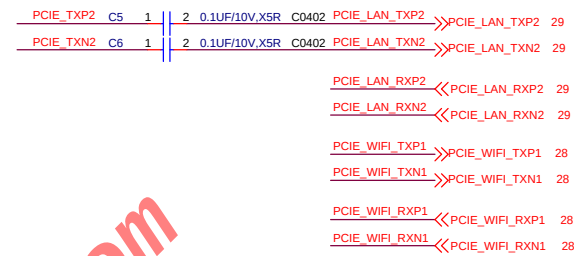
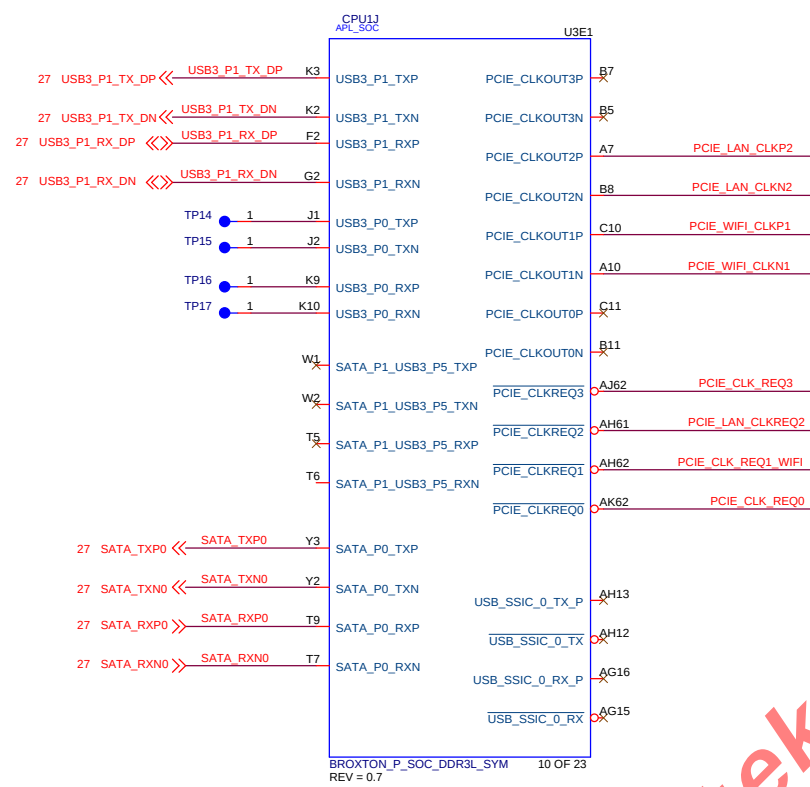
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14,25,26,28,29,30,31,32,33,34,38,39

+V1P8A  
+V1P8S  
+V3P3S





8,11,12,13,14,15,19,24,26,27,28,29,32,33,34,37,38,40 +V1P8A



8,10,12,13,14,15,19,24,26,27,28,29,32,33,34,37,38,40  
12,13,14,15,18,26,28,29,30,33,34,36,37,38,40

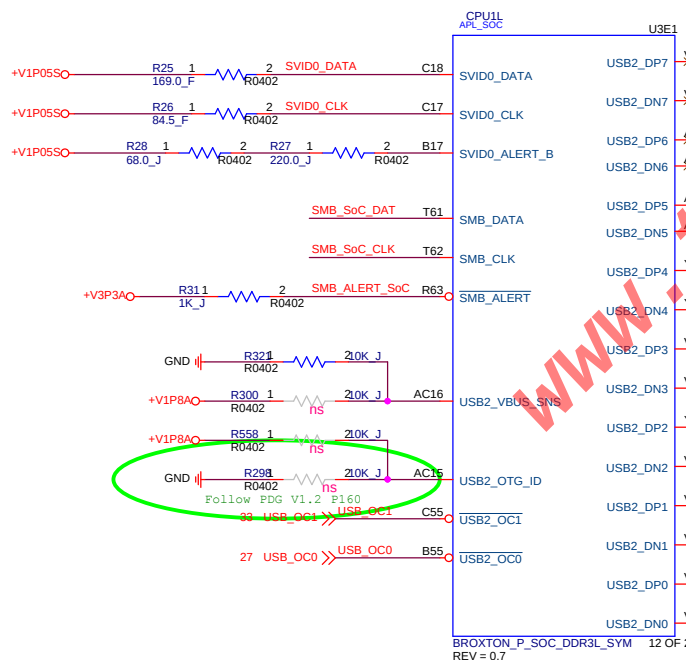
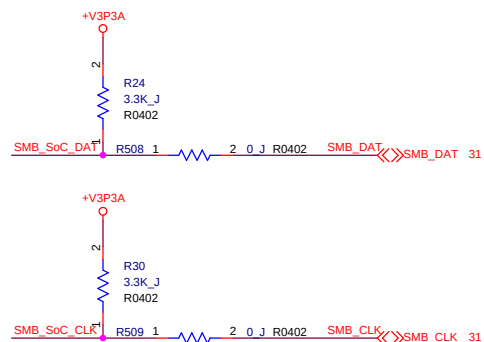
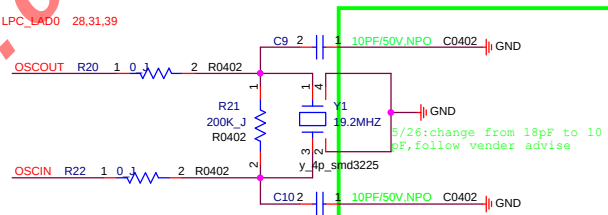
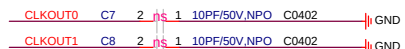
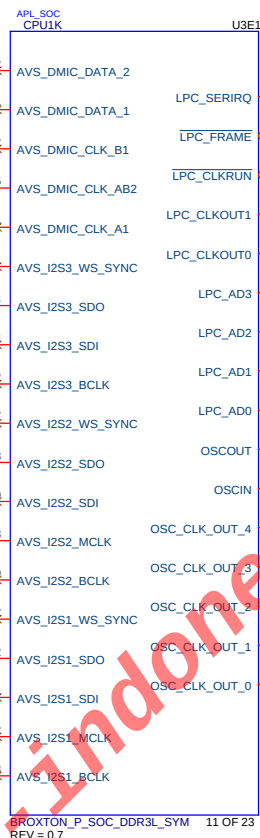
+V1P05S  
+V1P8A  
+V3P3A

24 GPIO\_82 << GPIO\_82

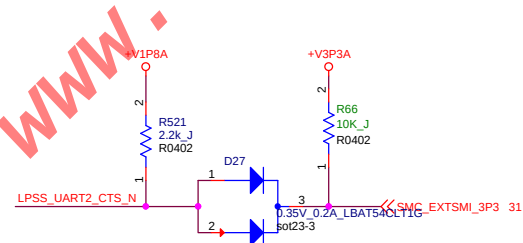
24 GPIO\_92 << GPIO\_92

24 GPIO\_88 << GPIO\_88

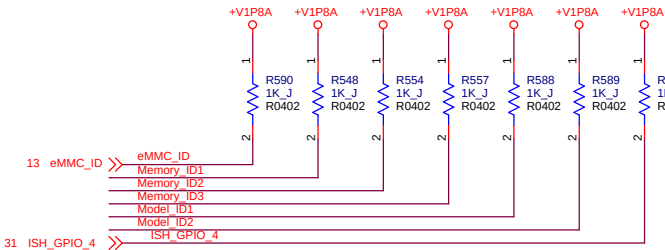
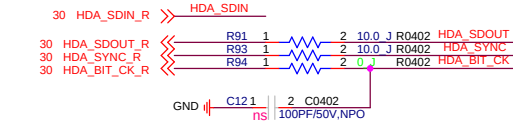
24 GPIO\_78 << GPIO\_78



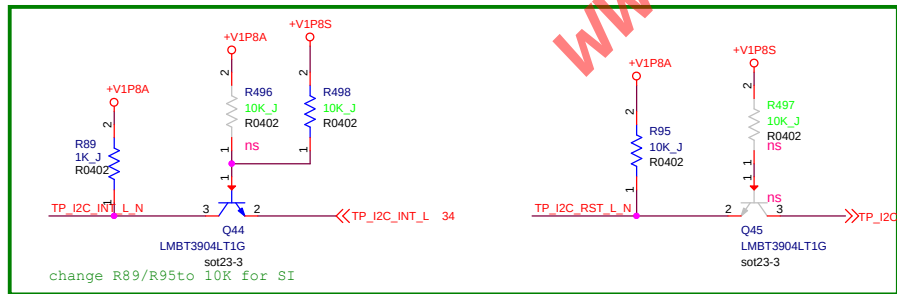
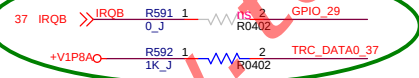




8,12,25,26,30,38 +VIP8S  
11,12,13,15,18,26,28,29,30,33,34,36,37,38,40 +V3P3A  
8,10,11,12,13,15,19,24,26,27,28,29,32,33,34,37,38,40 +VIP8A  
8,25,26,28,29,30,31,32,33,34,38,39 +V3P3S



| memory ID1(R548) | memory ID3(R557) | memory ID2(R554) | SDRAM configuration    |
|------------------|------------------|------------------|------------------------|
| GPIO_31          | GPIO_33          | GPIO_32          |                        |
| 1                | 1                | 1                | 4Gb x8                 |
| 1                | 1                | 0                | 8Gb x8                 |
| 0                | 1                | 0                | 8Gb x8 (DDP)           |
| 1                | 0                | 1                | 4Gb x4(channel 0)      |
| 1                | 0                | 0                | 8Gb x4(channel 0)      |
| 0                | 0                | 0                | 8Gb x4(channel 0 @DDP) |
| 0: DDP; 1: SDP   | 0: 单 1: 双        | 0: 8Gb 1: 4Gb    |                        |
| eMMC             | eMMC ID(R590)    |                  |                        |
| 上件               | 1                |                  |                        |
| 不上件              | 0                |                  |                        |
| Model Name       | model ID1(R588)  | model ID2(R589)  |                        |
| Yoga310          | 1                | 1                |                        |
| 310S-111AP       | 1                | 0                |                        |
| Flex4-1130       | 0                | 0                |                        |

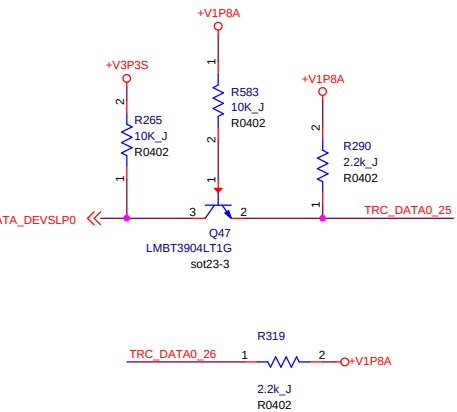
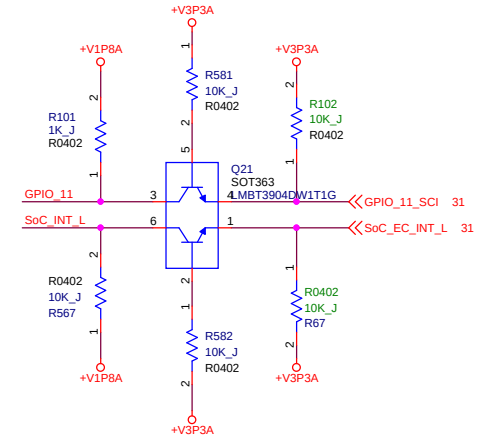
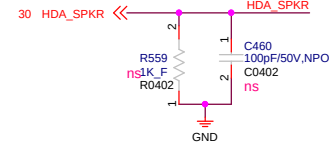


HDA\_SPKR AK57  
Model\_ID1 AM52  
Model\_ID2 AM55  
Gsensor\_INT0 Gsensor\_INT0\_N AM57  
Gsensor\_INT1 Gsensor\_INT1\_N AM49  
ISH\_GPIO\_4 AM51  
HDA\_SDOUT AM54  
ISH\_GPIO\_3  
TP33 1HDA\_SIN AK51  
ISH\_GPIO\_2  
HDA\_SYNC AK58  
ISH\_GPIO\_1  
HDA\_BIT\_CK AM48  
ISH\_GPIO\_0

Memory\_ID3 F34  
GPIO\_33  
Memory\_ID2 F35  
GPIO\_32  
Memory\_ID1 H34  
GPIO\_31  
TRC\_DATA0\_37 C37  
GPIO\_30  
TRC\_DATA0\_33 B37  
GPIO\_28  
TRC\_CLK2 C29  
GPIO\_27  
TRC\_DATA0\_27 C31  
GPIO\_26  
TRC\_DATA0\_26 C27  
GPIO\_25

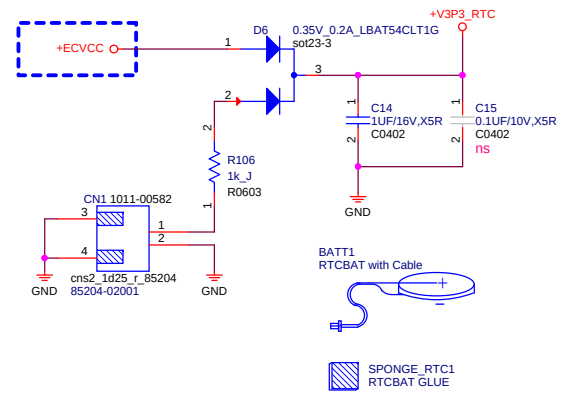
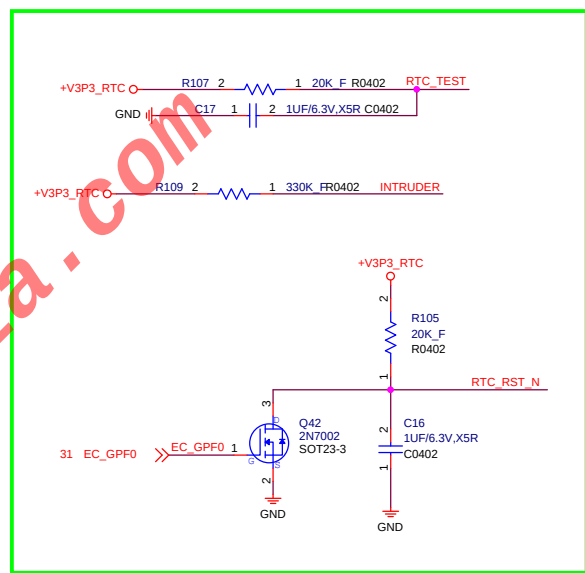
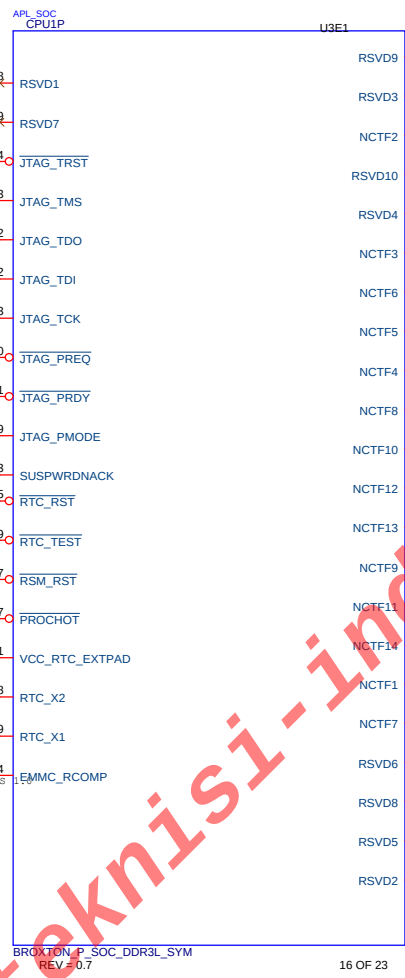
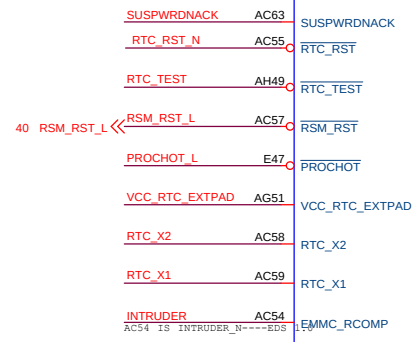
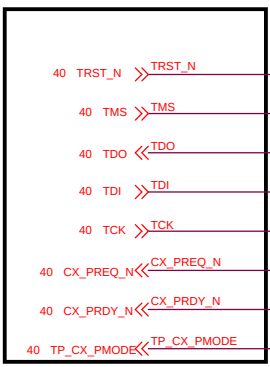
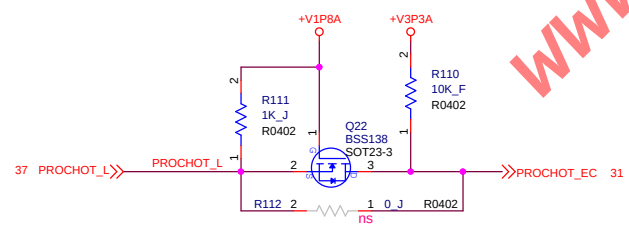
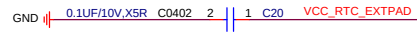
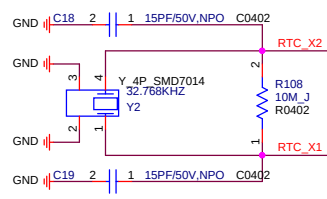
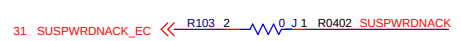
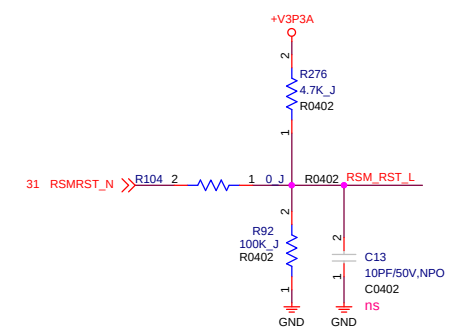
BROXTON\_P\_SOC\_DDR3L\_SYM  
REV = 0.7

U3E1  
GPIO\_18 J39 TRC\_CLK1 TRC\_CLK1 40  
GPIO\_24 C25 TRC\_DATA0\_25 TRC\_DATA0\_25 40  
GPIO\_23 B25 TRC\_DATA0\_24 TRC\_DATA0\_24 40  
GPIO\_22 A26 TRC\_DATA0\_23 TRC\_DATA0\_23 40  
GPIO\_21 C26 TRC\_DATA0\_22 TRC\_DATA0\_22 40  
GPIO\_20 B27 TRC\_DATA0\_21 TRC\_DATA0\_21 40  
GPIO\_19 C33 TRC\_DATA0\_20 TRC\_DATA0\_20 40  
GPIO\_17 A30 TP\_I2C\_INT\_L\_N TP\_I2C\_INT\_L\_N 40  
GPIO\_16 C35 TP\_I2C\_RST\_L\_N TP\_I2C\_RST\_L\_N 40  
GPIO\_15 C36 SoC\_INT\_L SoC\_INT\_L 40  
GPIO\_14 F39 TRC\_DATA0\_14 TRC\_DATA0\_14 40  
GPIO\_13 C38 TRC\_DATA0\_13 TRC\_DATA0\_13 40  
GPIO\_12 C30 TRC\_DATA0\_12 TRC\_DATA0\_12 40  
GPIO\_11 E39 TRC\_DATA0\_11 TRC\_DATA0\_11 40  
GPIO\_10 C34 GPIO\_11 GPIO\_11 40  
GPIO\_9 L39 TRC\_DATA0\_9 TRC\_DATA0\_9 40  
GPIO\_8 A38 TRC\_CLK0 TRC\_CLK0 40  
GPIO\_7 B29 TRC\_DATA0\_8 TRC\_DATA0\_8 40  
GPIO\_6 H39 TRC\_DATA0\_7 TRC\_DATA0\_7 40  
GPIO\_5 B31 TRC\_DATA0\_6 TRC\_DATA0\_6 40  
GPIO\_4 A34 TRC\_DATA0\_5 TRC\_DATA0\_5 40  
GPIO\_3 B35 TRC\_DATA0\_4 TRC\_DATA0\_4 40  
GPIO\_2 B39 TRC\_DATA0\_3 TRC\_DATA0\_3 40  
GPIO\_1 C39 TRC\_DATA0\_2 TRC\_DATA0\_2 40  
GPIO\_0 B33 TRC\_DATA0\_1 TRC\_DATA0\_1 40

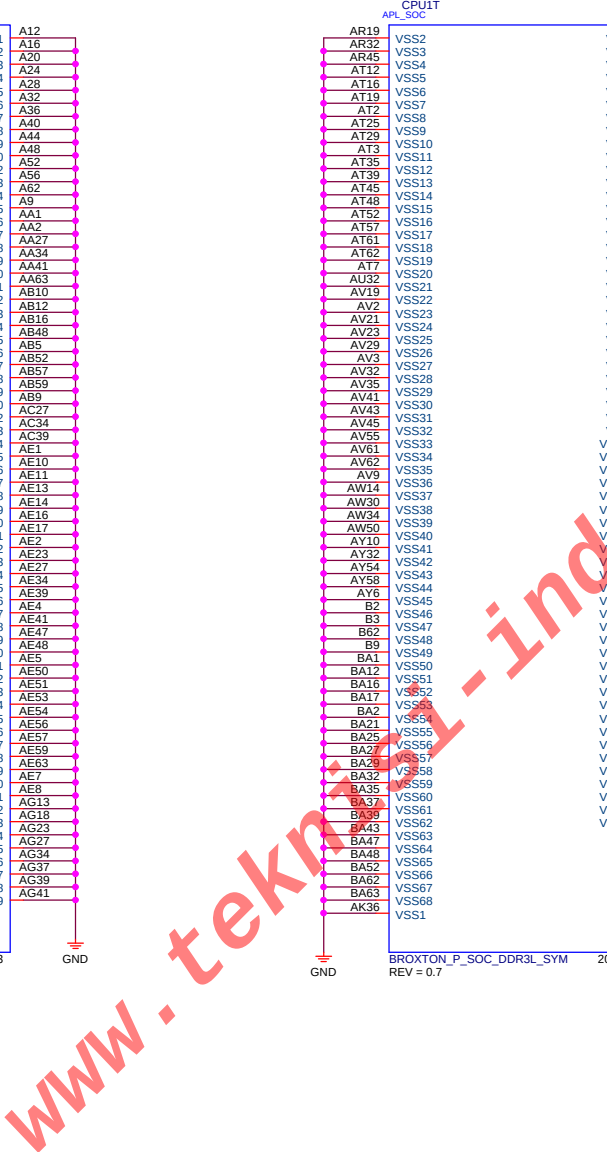
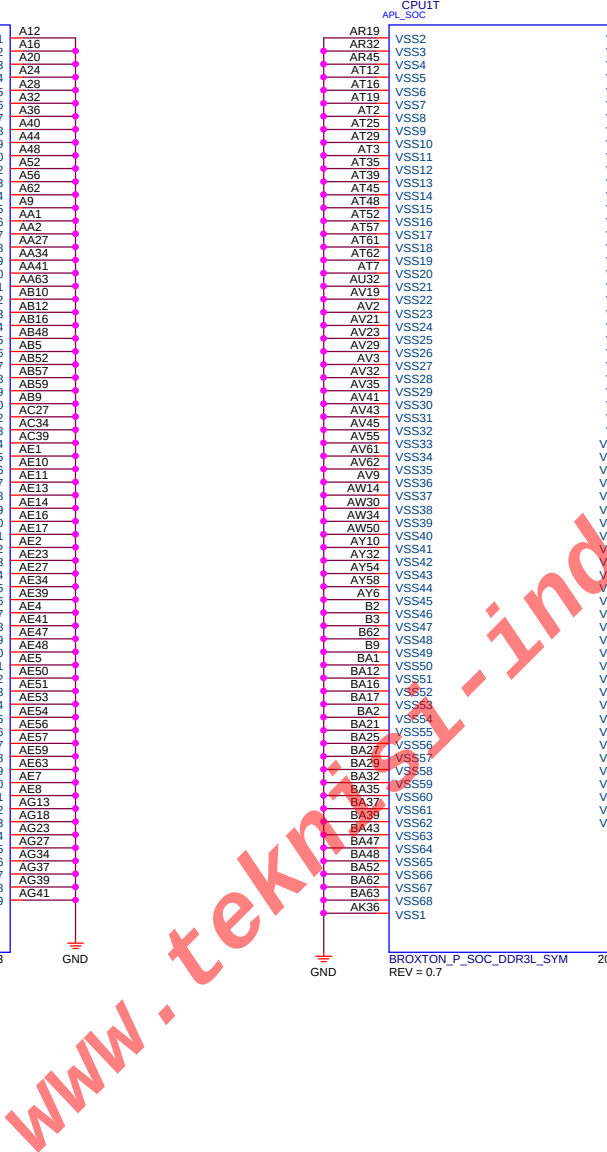
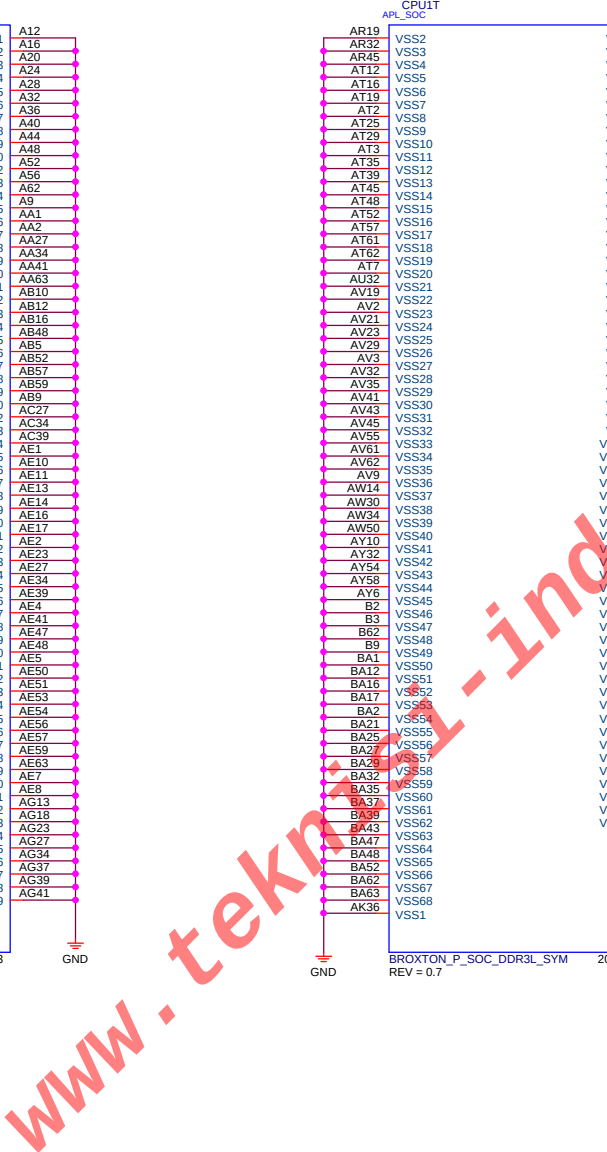


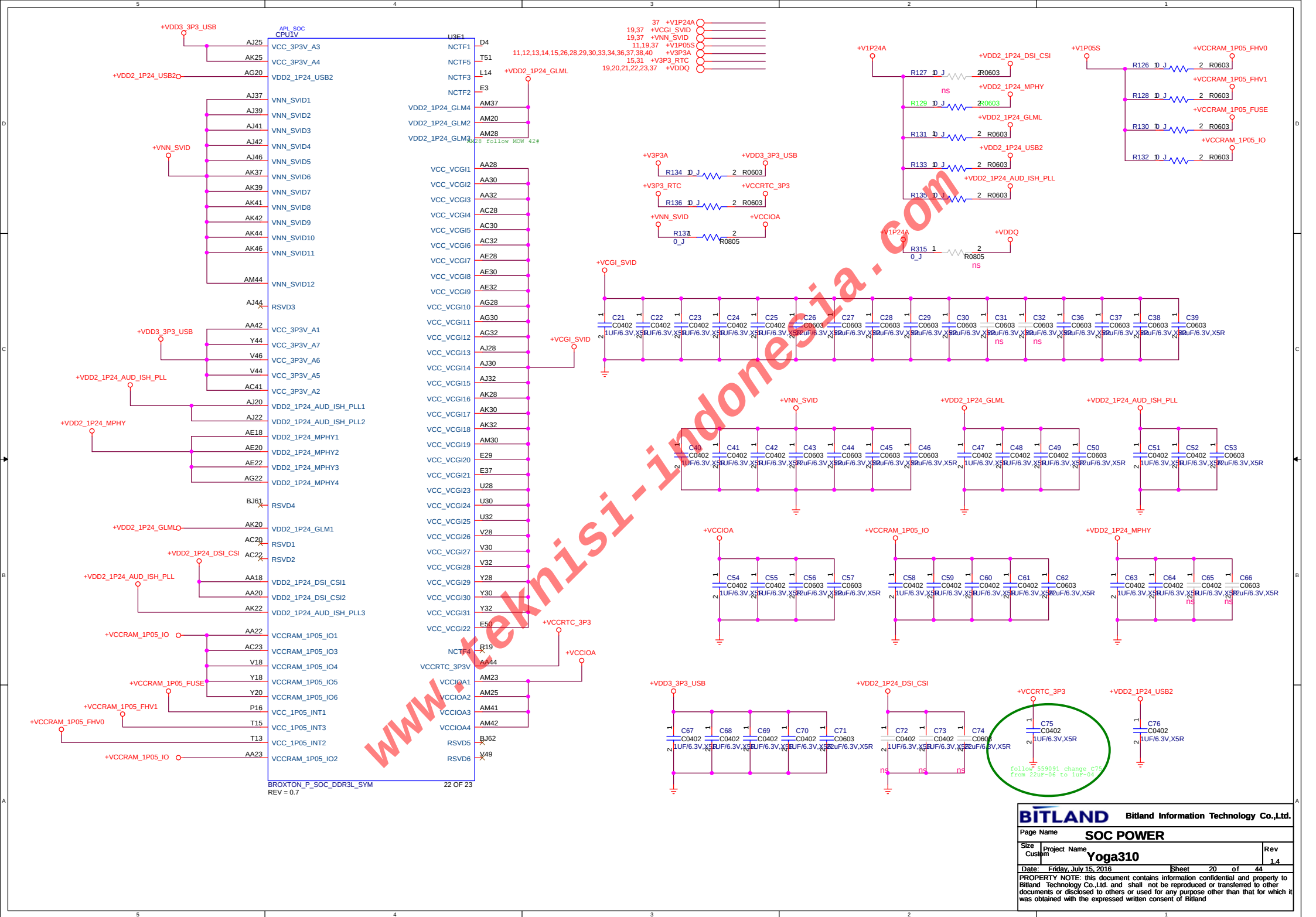
|                                                                                                                                                                                                                                                                                                                 |                |       |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------|--|
| <b>BITLAND</b> Bitland Information Technology Co.,Ltd.                                                                                                                                                                                                                                                          |                |       |  |
| Page Name <b>SOC HDA/GPIO</b>                                                                                                                                                                                                                                                                                   |                |       |  |
| Size                                                                                                                                                                                                                                                                                                            | Project Name   | Rev   |  |
| Custom                                                                                                                                                                                                                                                                                                          | <b>Yoga310</b> | 1.4   |  |
| Date: Friday, July 15, 2016                                                                                                                                                                                                                                                                                     | Sheet 16       | of 44 |  |
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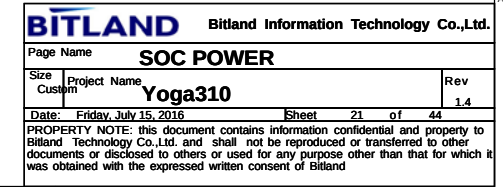
11,12,13,14,18,26,28,29,30,33,34,36,37,38,40 +V3P3A  
 8,10,11,12,13,14,19,24,26,27,28,29,32,33,34,37,38,40 +V1P8A  
 27,31,32,33,35,36 +ECVCC  
 18,31 +V3P3\_RTC

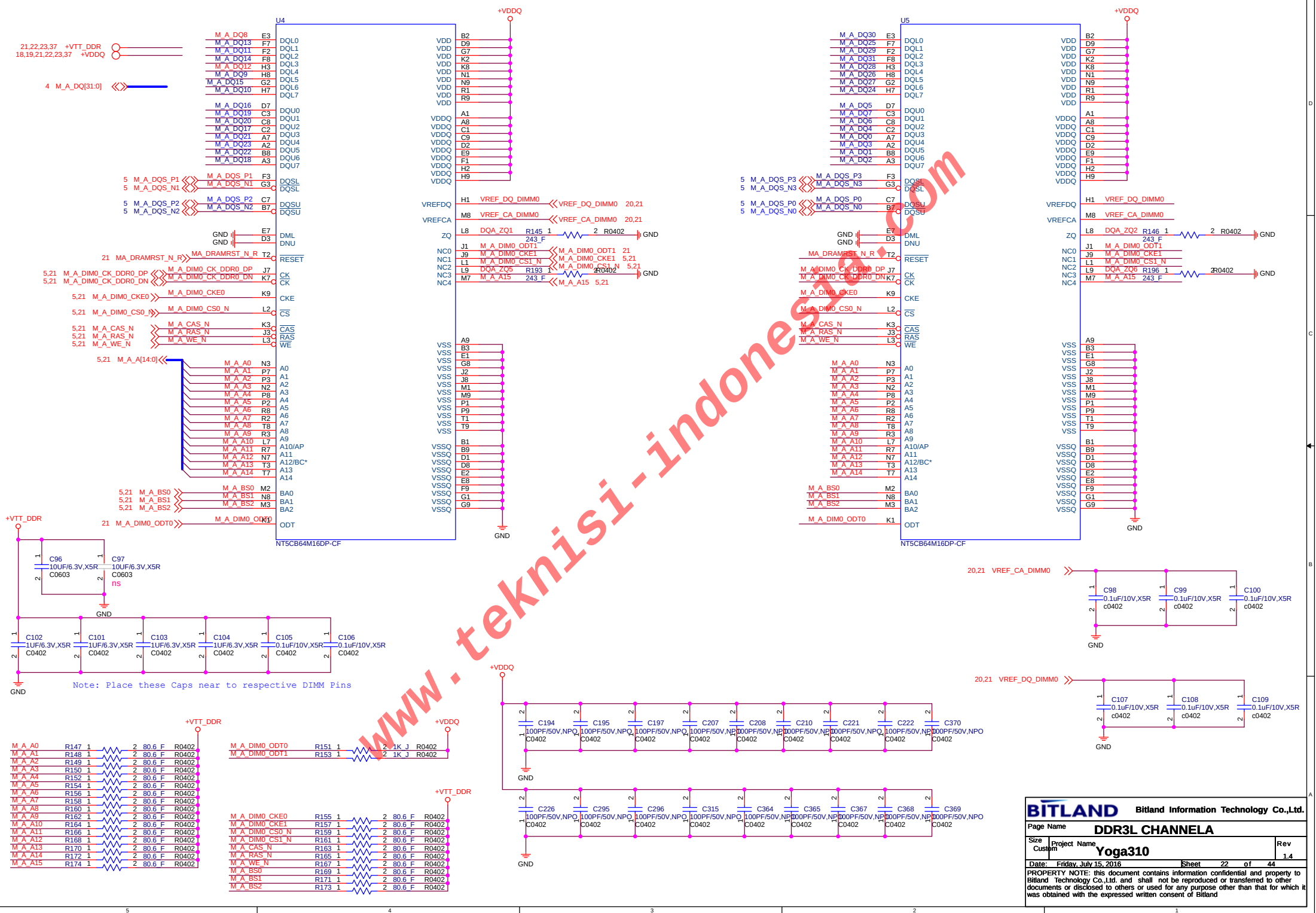


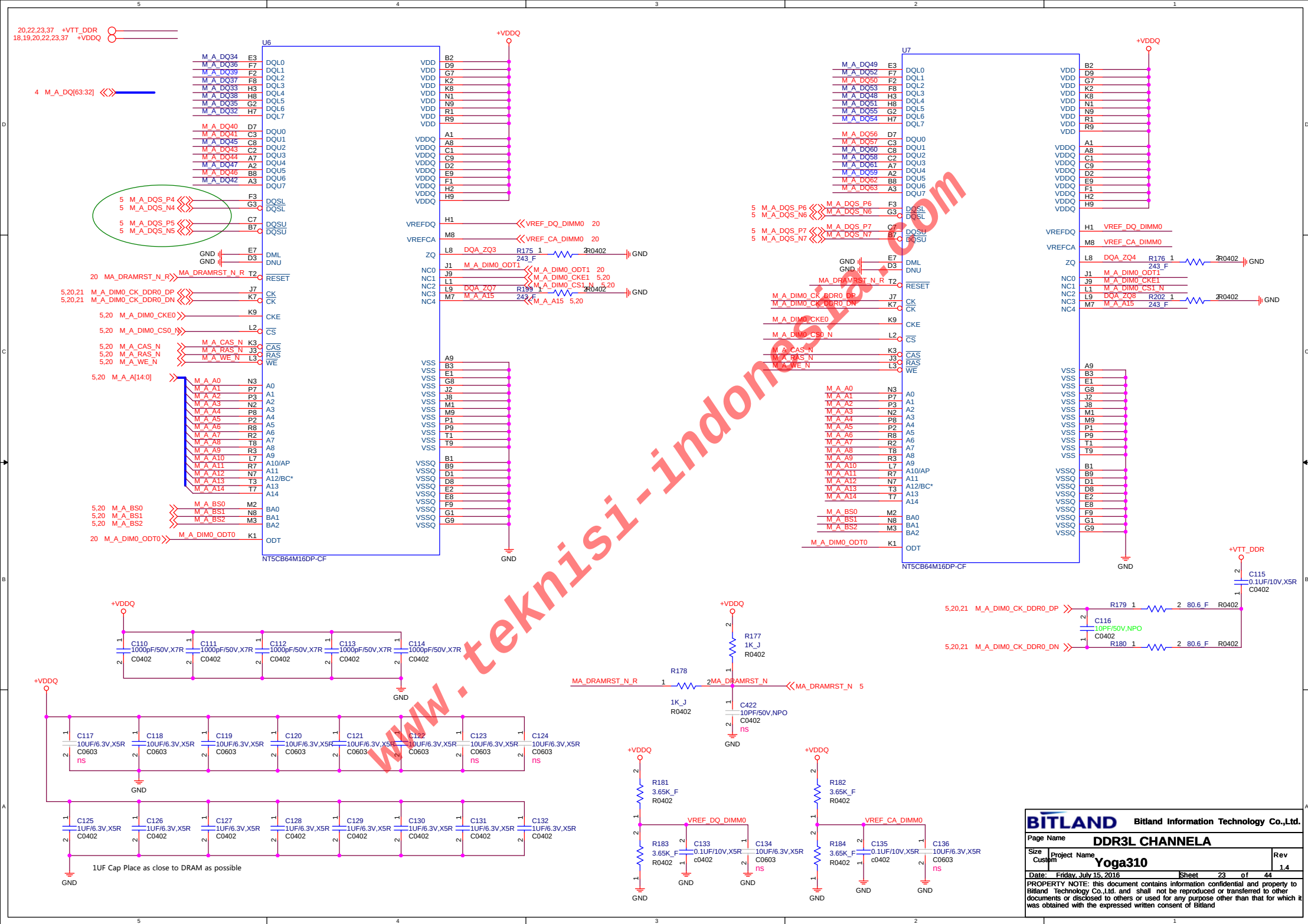


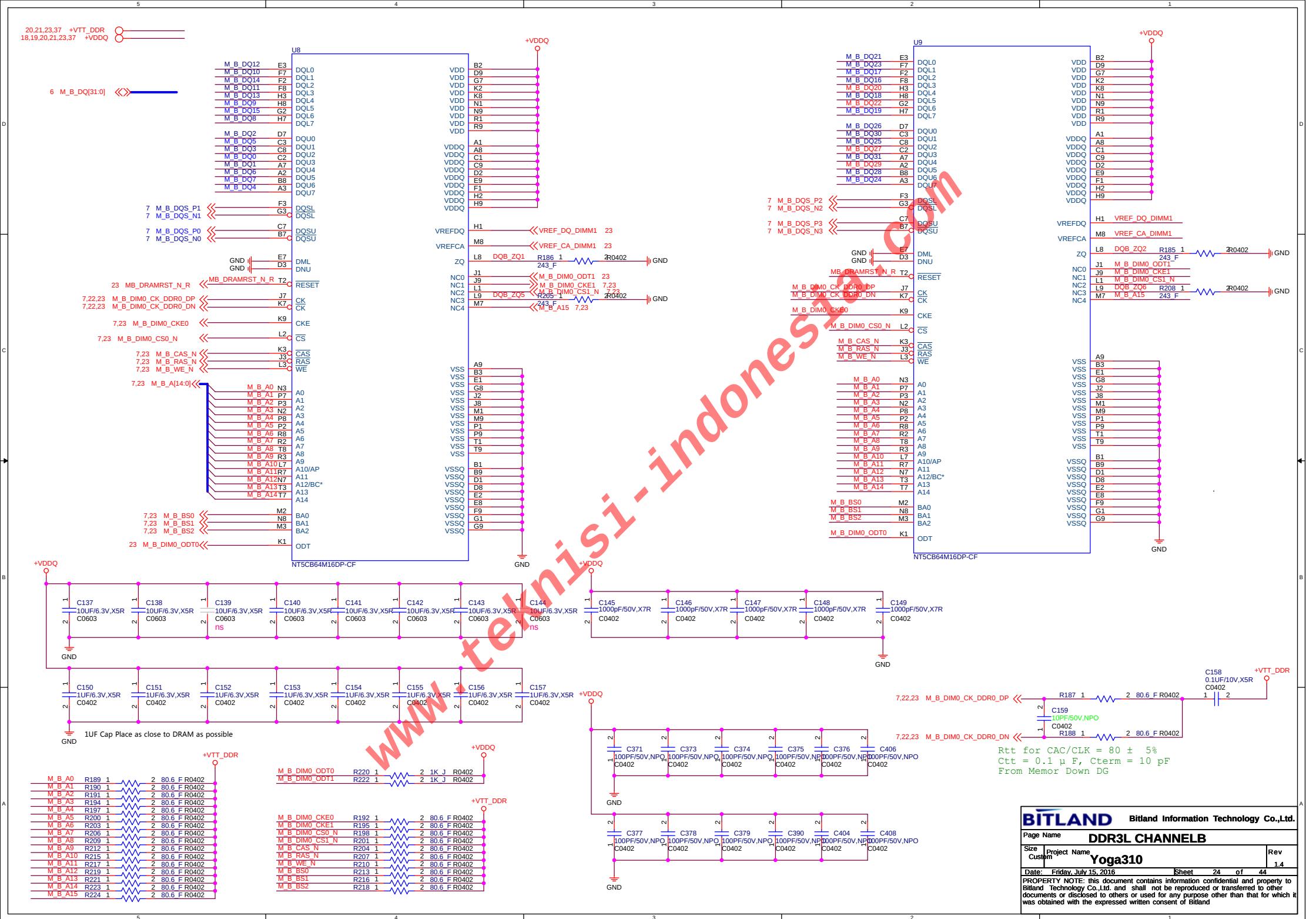














|       |         |         |       |          |          |    |          |          |
|-------|---------|---------|-------|----------|----------|----|----------|----------|
| 16    | GPIO_34 | GPIO_34 | 12    | GPIO_104 | GPIO_104 | 13 | GPIO_40  | GPIO_40  |
| 16    | GPIO_35 | GPIO_35 | 12    | GPIO_105 | GPIO_105 | 11 | GPIO_82  | GPIO_82  |
| 16    | GPIO_36 | GPIO_36 | 12    | GPIO_106 | GPIO_106 | 11 | GPIO_88  | GPIO_88  |
| 13    | GPIO_39 | GPIO_39 | 12,40 | GPIO_123 | GPIO_123 | 11 | GPIO_92  | GPIO_92  |
| 13,40 | GPIO_43 | GPIO_43 | 12,40 | GPIO_121 | GPIO_121 | 12 | GPIO_113 | GPIO_113 |
| 13    | GPIO_44 | GPIO_44 | 12    | GPIO_122 | GPIO_122 | 12 | GPIO_112 | GPIO_112 |
| 13    | GPIO_47 | GPIO_47 | 12    | GPIO_120 | GPIO_120 | 12 | GPIO_111 | GPIO_111 |
| 11    | GPIO_78 | GPIO_78 | 12    | GPIO_118 | GPIO_118 | 12 | GPIO_110 | GPIO_110 |
|       |         |         | 12    | GPIO_117 | GPIO_117 | 12 | GPIO_110 | GPIO_110 |
|       |         |         |       |          |          | 13 | GPIO_48  | GPIO_48  |

8,10,11,12,13,14,15,19,26,27,28,29,32,33,34,37,38,40 +V1P8A

1 = Do not boot from SPI  
0 = Boot from SPI(default)

SMBus 1.8V/3.3V mode select  
1=buffers set to 1.8V mode  
0=buffers set to 3.3V mode(default)

1 = Force  
0 = Do not force (default)  
Note: This strap is a recovery strap for corrupted FW image.

SMBus No Re-Boot  
1 = Enable  
0 = Disable (default)  
Note: Platforms should strap this LOW. Functionality is handled by the PMC.

VCC\_1P24V\_1P35V\_A voltage selection  
1 = 1.35V  
0 = 1.24V (default)  
Note: This strap will only be used for B-step. For A-step this rails should only be set at 1.24V

Enable CSE ROM Bypass  
1 = enable bypass  
0 = disable bypass (default)  
Note: This strap tells CSE to bypass Read-Only Memory (ROM) that it has on SoC.

those straps is pulled HIGH when RSM\_RST\_N de-asserts for normal platform operation.

Allow eMMC as a boot source  
1=enable (default)  
0=disable (default)  
Note: If platform is using SPI as the boot device then please provide a pull-down for this strap to disable eMMC.

Allow SPI as a boot source  
1=enable (default)  
0=disable (default)  
Note: If platform is using eMMC as boot device then please provide a pull down for this strap to disable SPI.

PMU 1.8V/3.3V mode select  
1=buffers set to 1.8V mode  
0=buffers set to 3.3V mode (default)

LP1 1.8V/3.3V mode select  
1=buffers set to 1.8V mode  
0=buffers set to 3.3V mode(default)

Top swap override  
1 = Enable  
0 = Disable (default)  
Note: This strap enables platform to change where the core will look for BIOS code for a SPI boot only.

Flash Descriptor Override  
0 = No Override (Normal Operation)  
1 = Override  
Note: This strap enables the platform to override security features in the SPI.

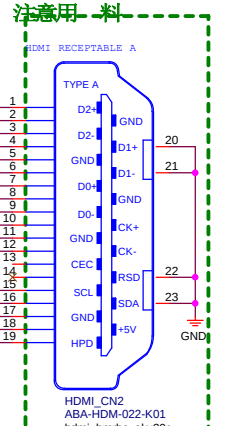
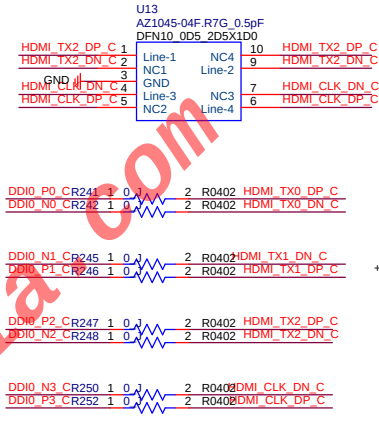
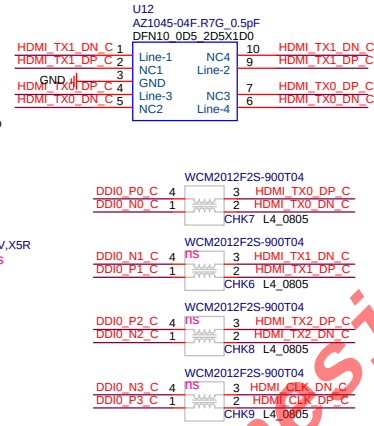
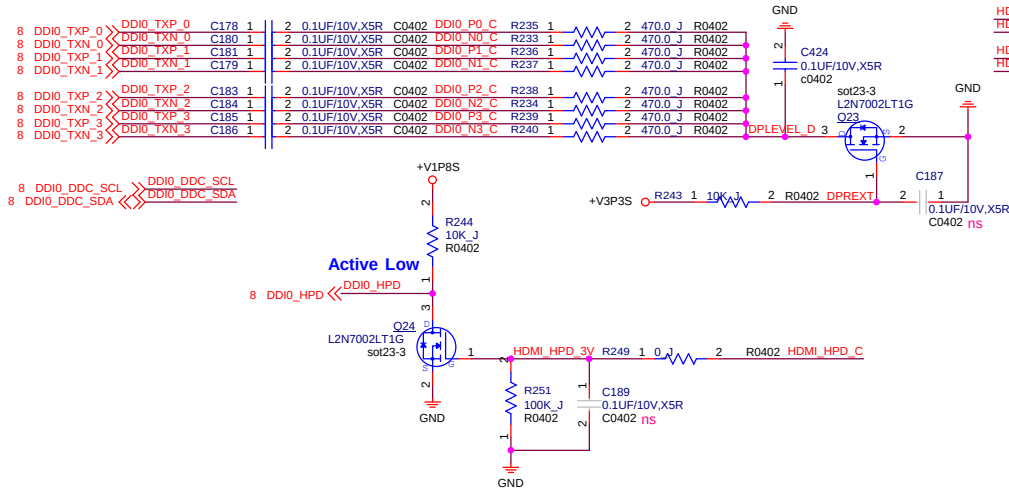
this strap is always pulled low for normal platform operation.

those straps is pulled LOW when RSM\_RST\_N de-asserts for normal platform operation.

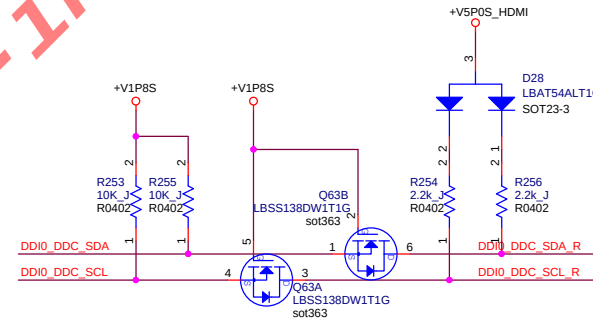
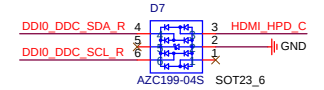
|                                                                                                                                                                                                                                                                                                                 |                             |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|
| <b>BITLAND</b> Bitland Information Technology Co.,Ltd.                                                                                                                                                                                                                                                          |                             |                |
| Page Name <b>HW STRAP</b>                                                                                                                                                                                                                                                                                       |                             |                |
| Size Custom                                                                                                                                                                                                                                                                                                     | Project Name <b>Yoga310</b> | Rev <b>1.4</b> |
| Date: Friday, July 15, 2016                                                                                                                                                                                                                                                                                     | Sheet 26 of 44              |                |
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## HDMI

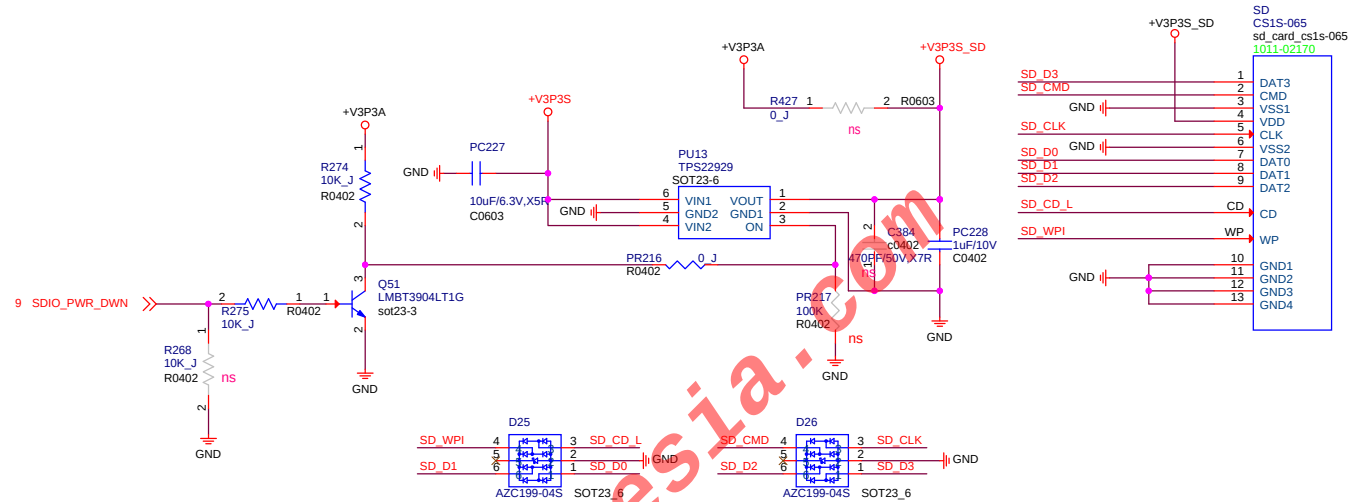
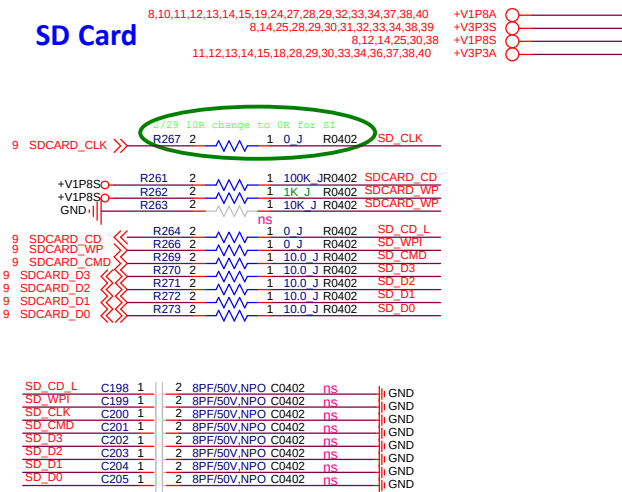
8,14,26,28,29,30,31,32,33,34,38,39 +V3P3S  
8,12,14,26,30,38 +V1P8S  
27,30,38 +V5P0S



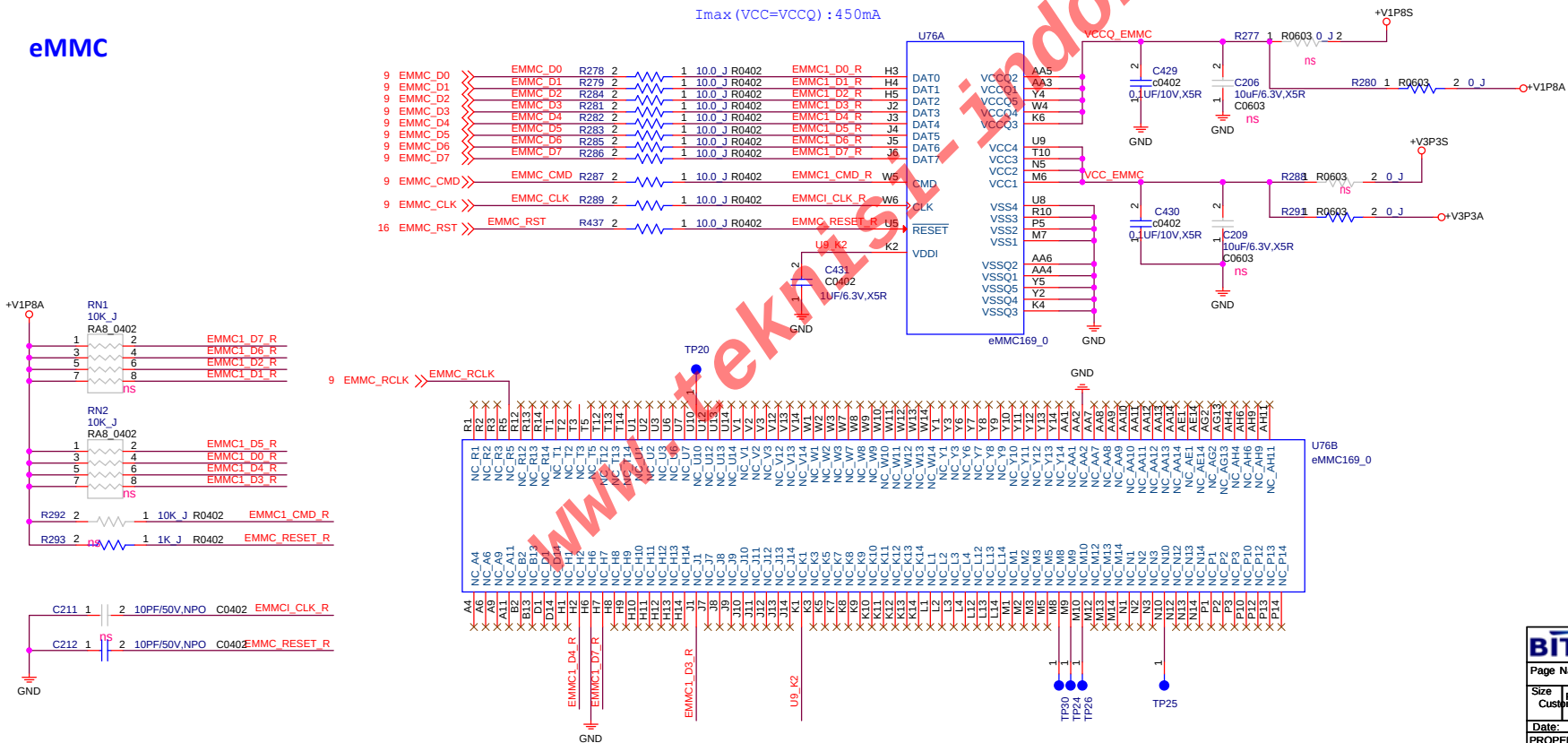
DDIO\_DDC\_SDA\_R C190 2 1 10PF/50V,NPOC0402 ns GND  
DDIO\_DDC\_SCL\_R C191 2 1 10PF/50V,NPOC0402 ns GND

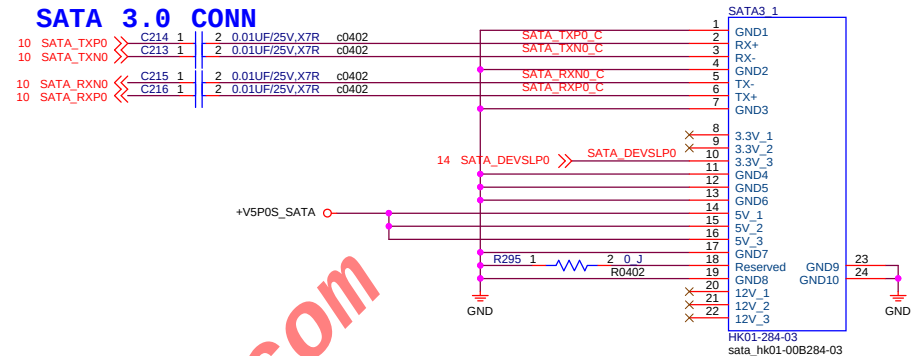
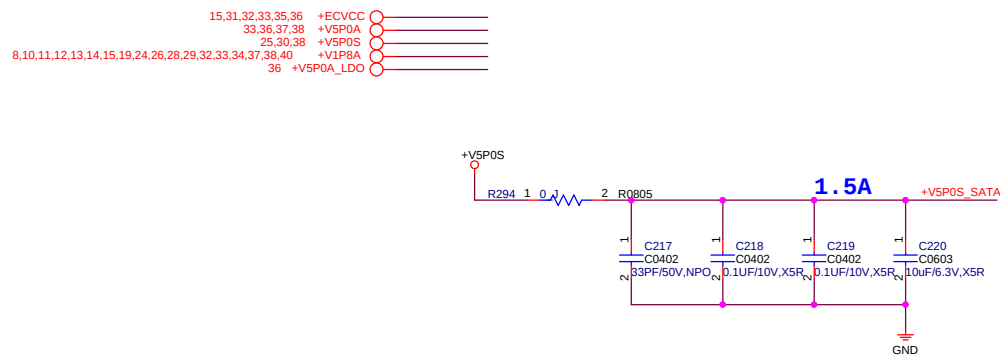


## SD Card

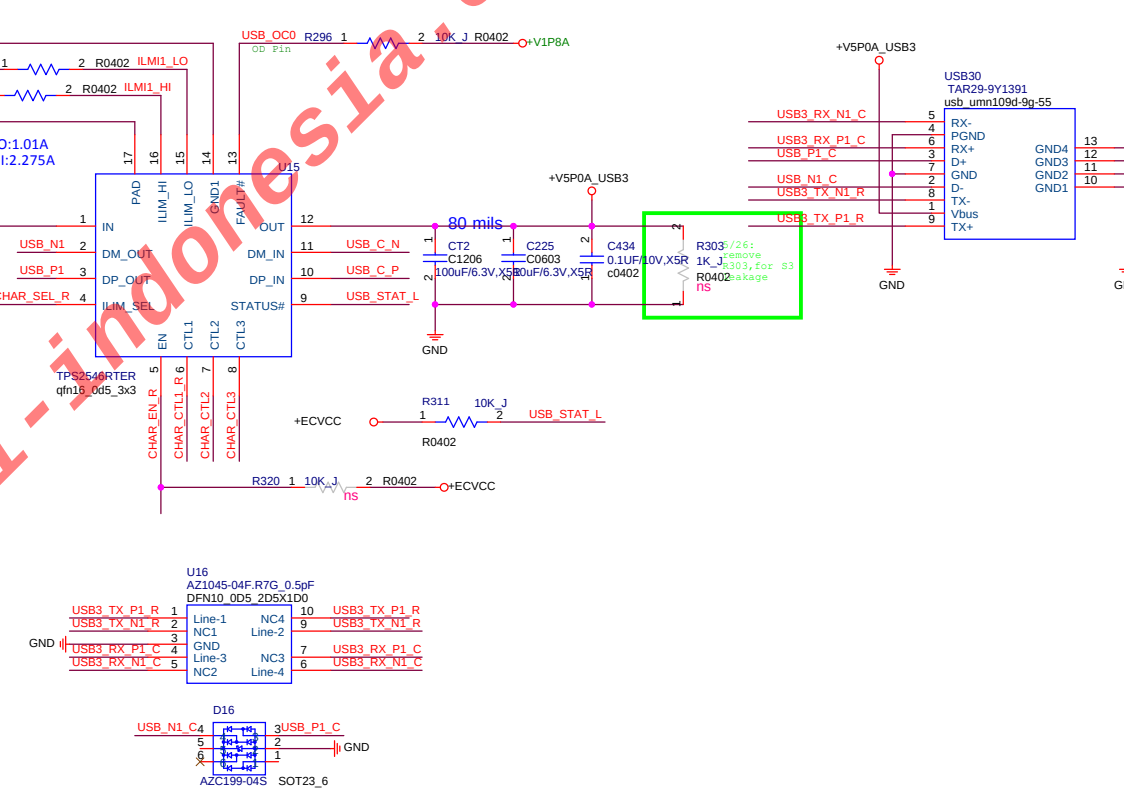
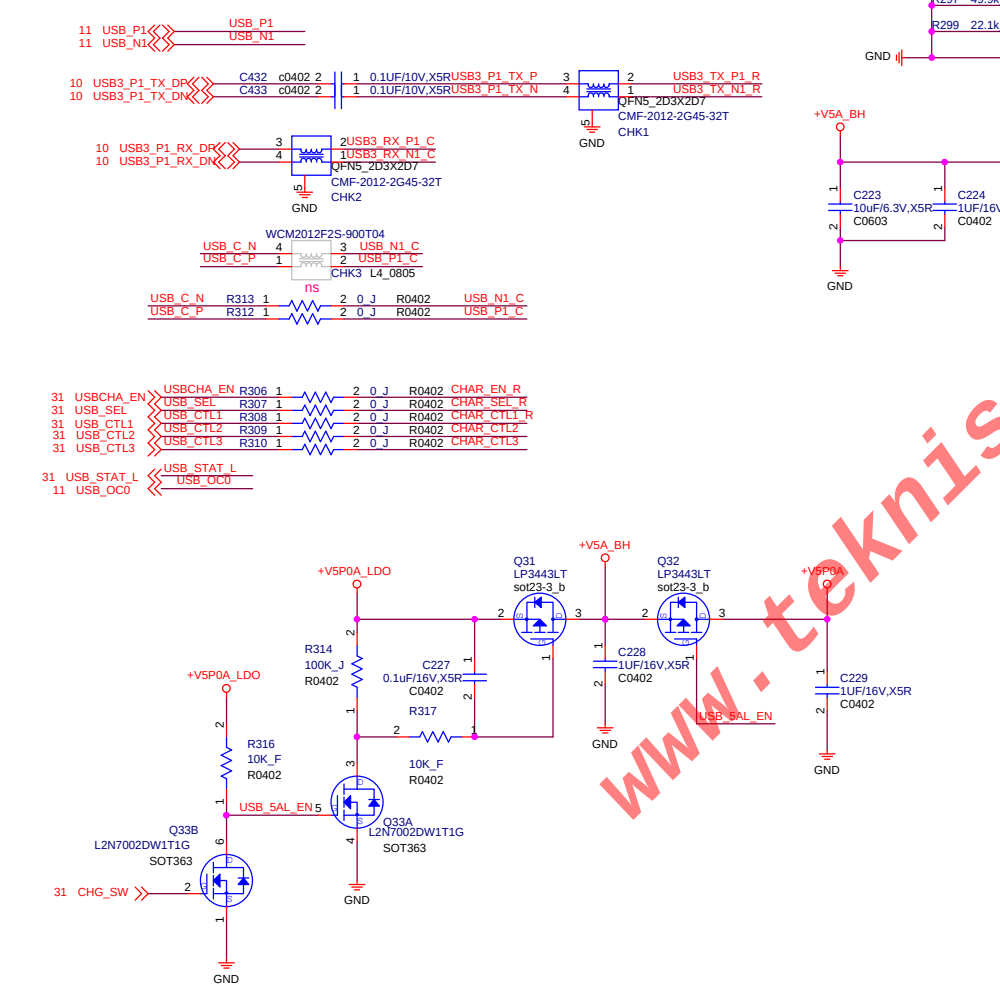


**eMMC**





## USB3.0 With Charge



## TPS2546 Control Mode (For Adapter)

| Input Logic Level |        |        |            | Control Mode  |                               |
|-------------------|--------|--------|------------|---------------|-------------------------------|
| INPUT1*           | INPUT2 | INPUT3 | ILIM_SEL * | System Status | Charging Mode                 |
| 0                 | 0      | 1      | 1          | S4/S5         | DCP(Dedicated Charging Port)  |
| 0                 | 1      | 1      | 1          | S3            | DCP(Dedicated Charging Port)  |
| 1                 | 1      | 1      | 1          | S0            | CDP(Charging Downstream Port) |

**BITLEND** Bitland Information Technology Co.,Ltd.

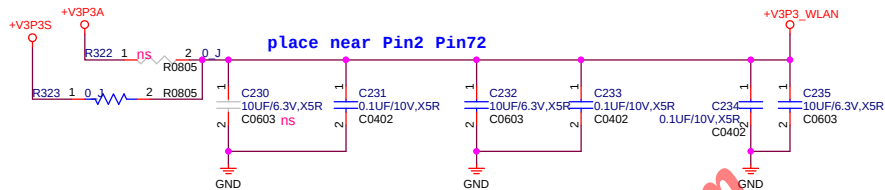
Page Name **SATA3.0&USB3.0**

Size Custom **Yoga310** Rev 1.4

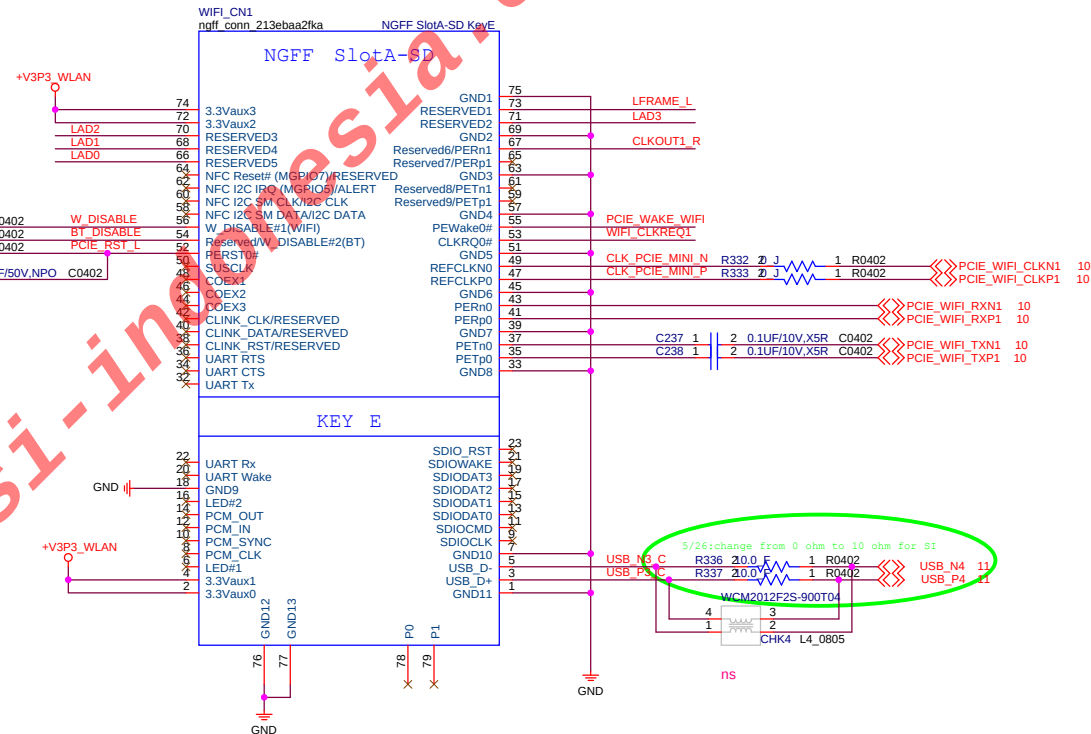
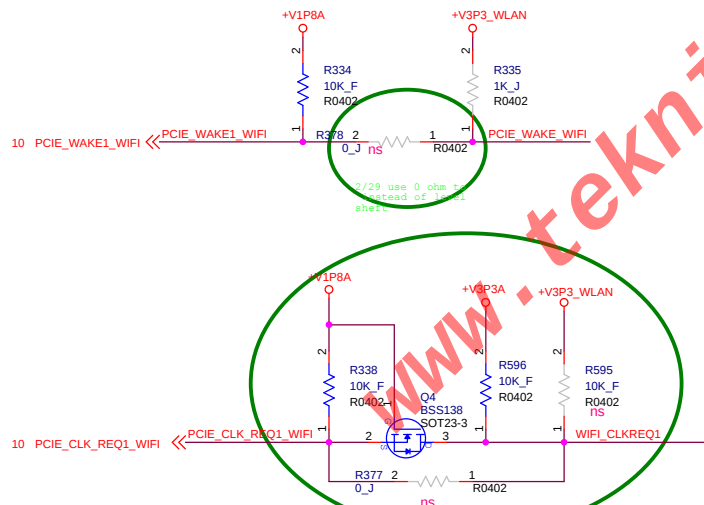
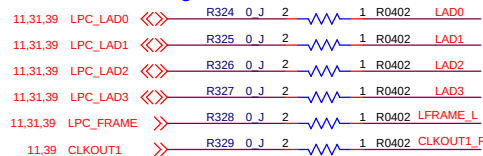
Date: Friday, July 15, 2016 Sheet 29 of 44

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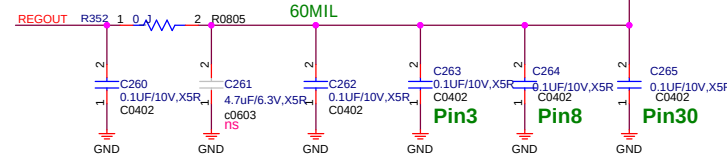
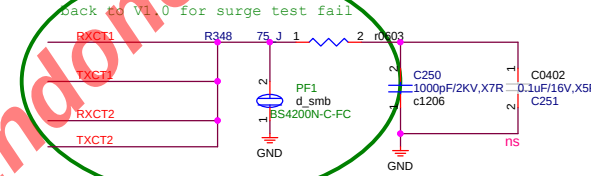
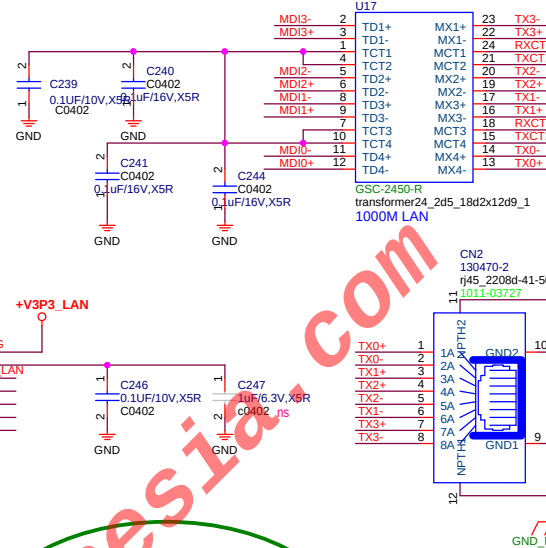
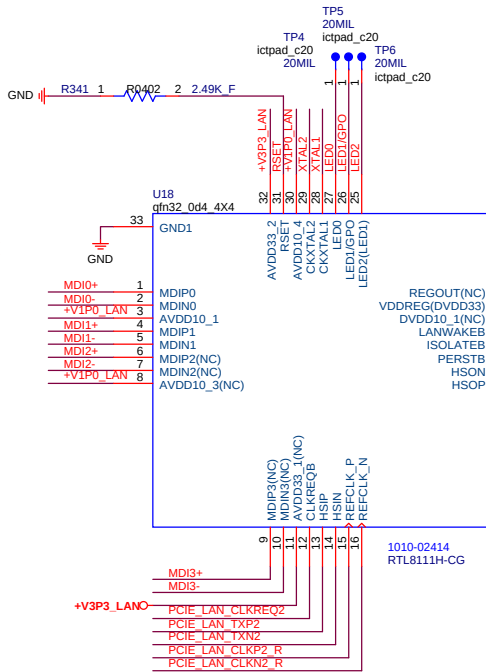
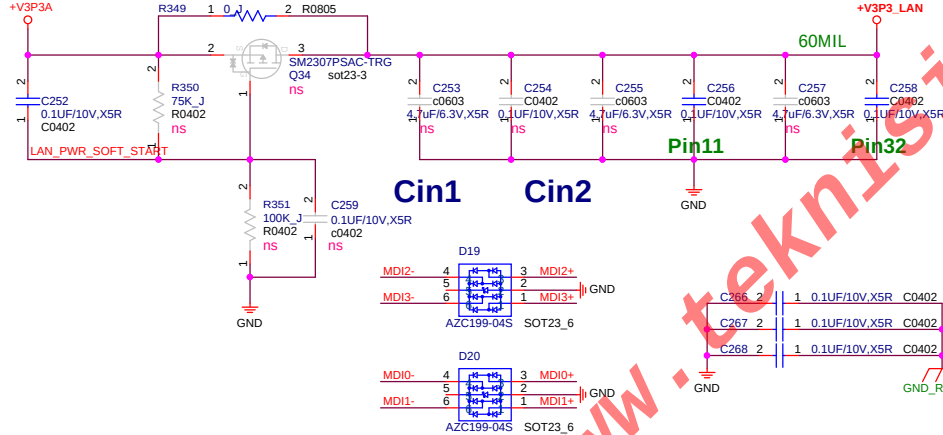
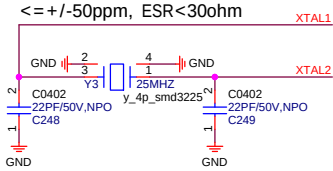
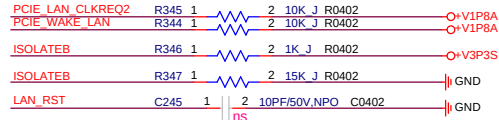
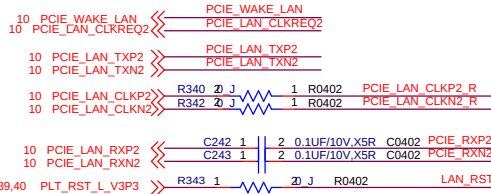
8,10,11,12,13,14,15,19,24,26,27,29,32,33,34,37,38,40 +V1P8A  
8,14,25,26,29,30,31,32,33,34,38,39 +V3P3S  
11,12,13,14,15,18,26,29,30,33,34,36,37,38,40 +V3P3A



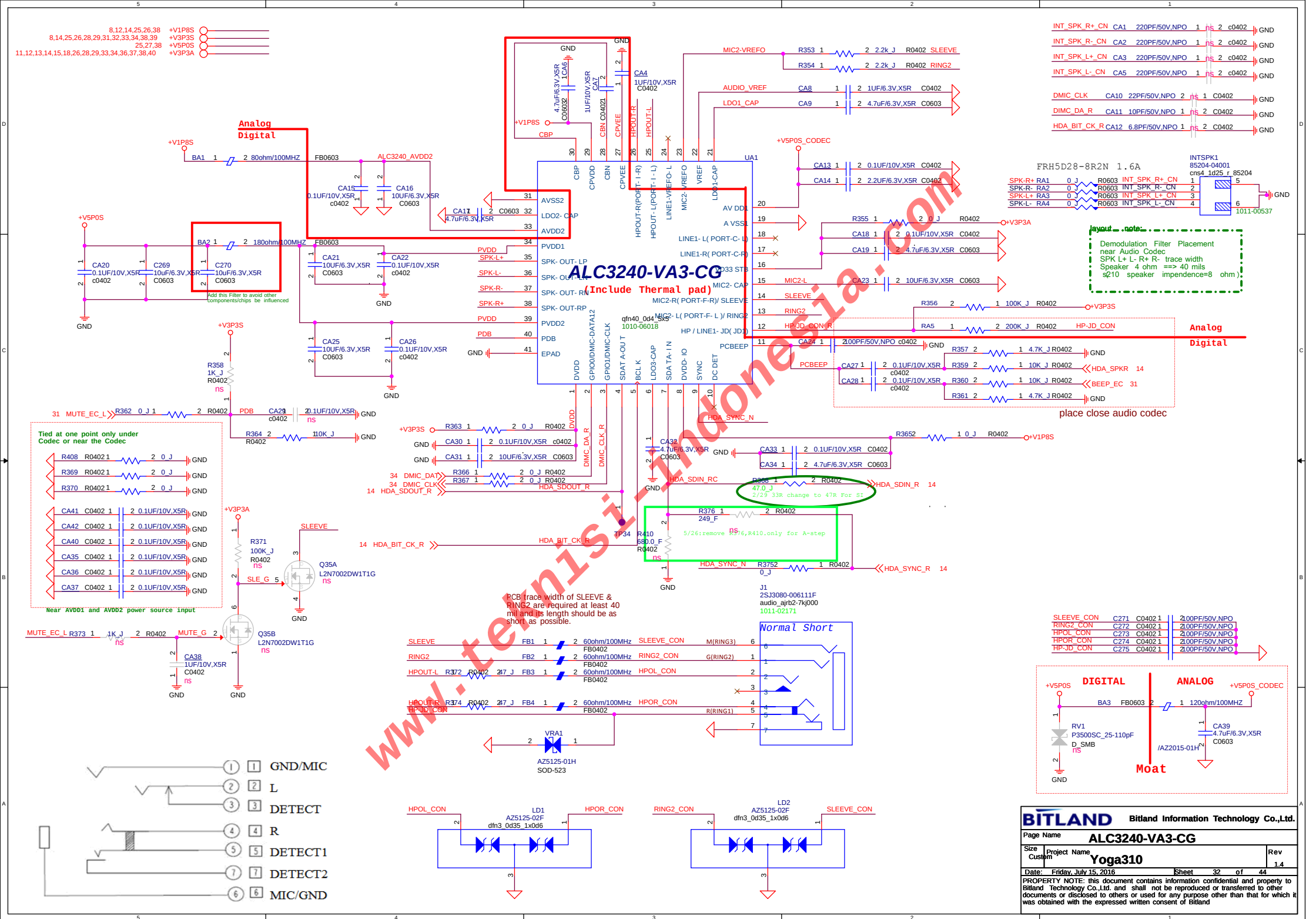
### For Debug Card

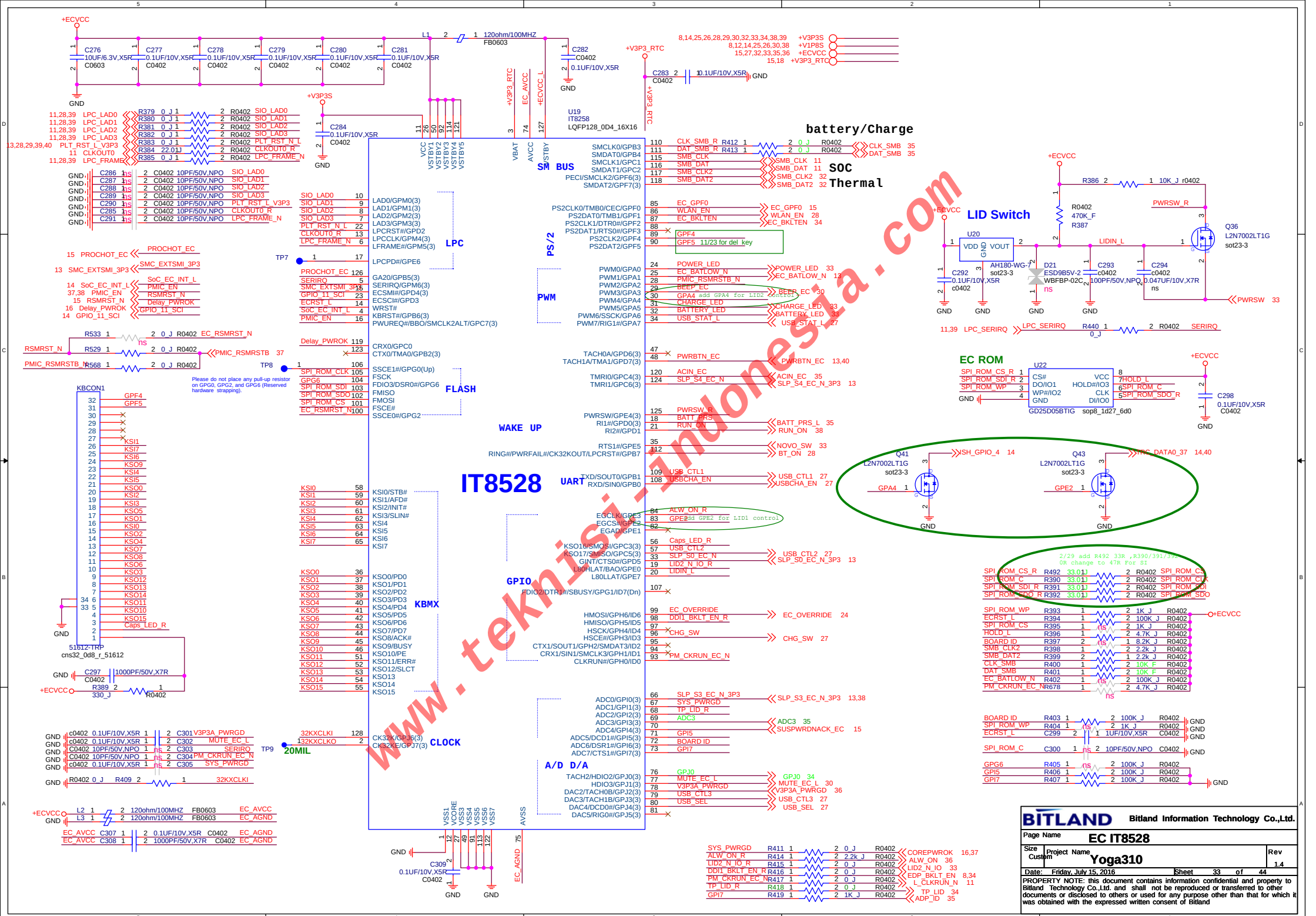


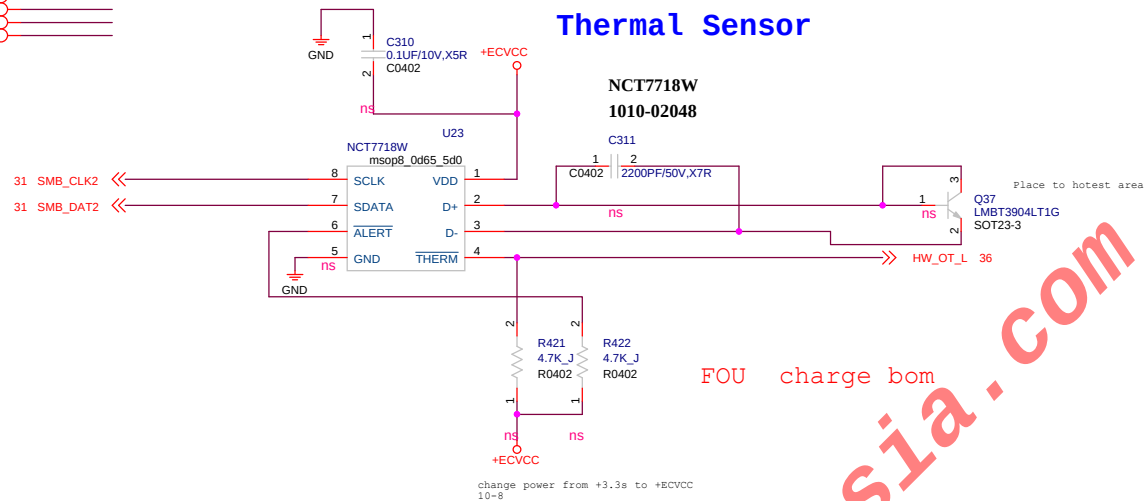
8,10,11,12,13,14,15,19,24,26,27,28,32,33,34,37,38,40 +VIP8A  
8,14,25,26,28,30,31,32,33,34,38,39 +V3P3S  
11,12,13,14,15,18,26,28,30,33,34,36,37,38,40 +V3P3A



RTL8111GUL&RTL8106EUL do not support LDO mode  
Cin1&Cin2 should be placed on the path from 33VAUX to VDDREG/VDD33



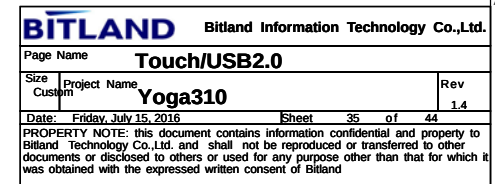


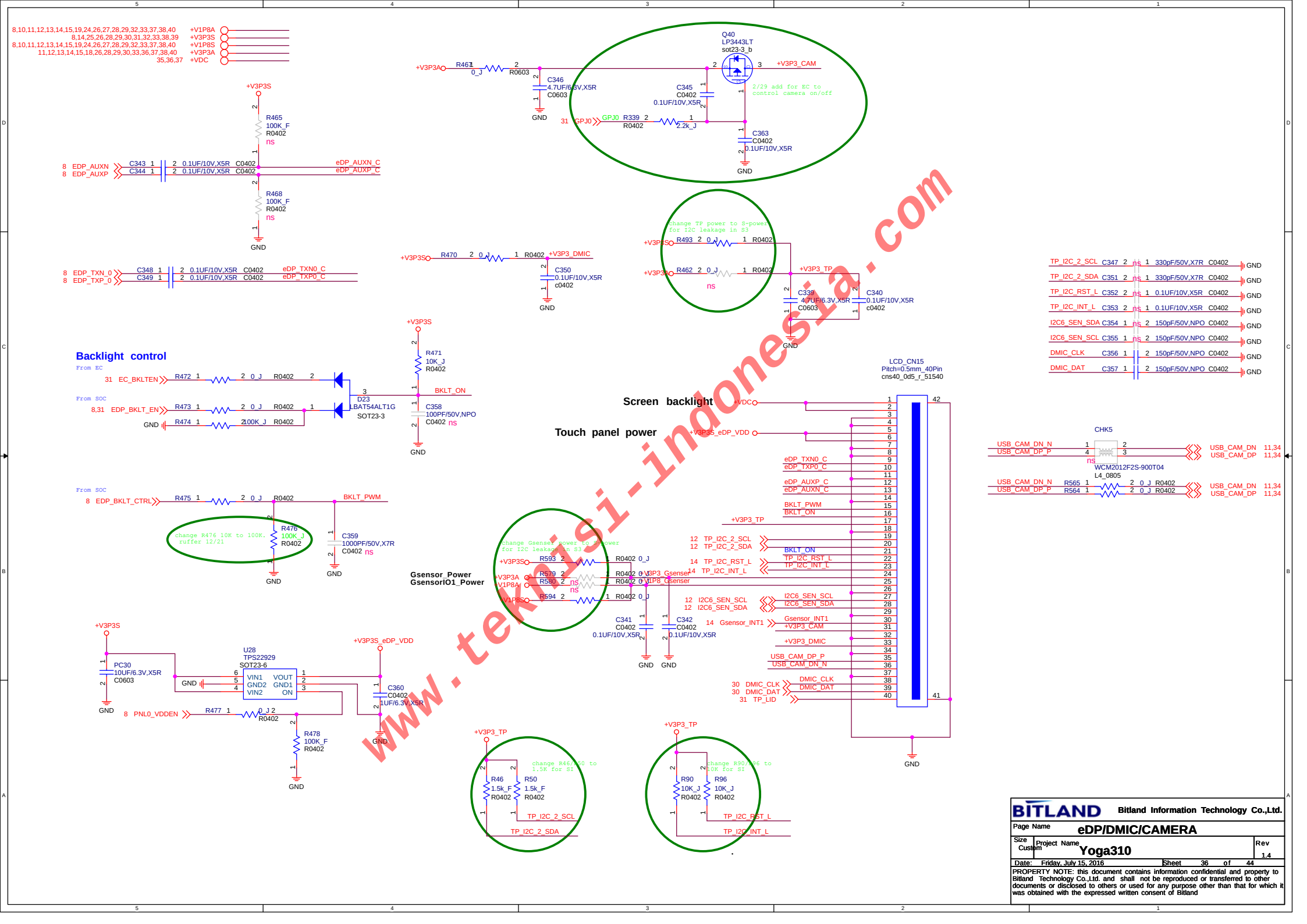


Note:  
The first pin of G-sensor must be  
place on the lower left corner of PCB.

3D G-Sensor

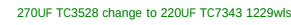
|                                                                                                                                                                                                                                                                                                                  |                       |                                          |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------|----------|
|                                                                                                                                                                                                                             |                       | Bitland Information Technology Co., Ltd. |          |
| Page Name                                                                                                                                                                                                                                                                                                        |                       | <b>GSENSOR THERMAL</b>                   |          |
| Size                                                                                                                                                                                                                                                                                                             | Project Name          | Rev                                      |          |
| Custom                                                                                                                                                                                                                                                                                                           | <b>Yoga310</b>        | 1.4                                      |          |
| Date:                                                                                                                                                                                                                                                                                                            | Friday, July 15, 2016 | Sheet                                    | 34 of 44 |
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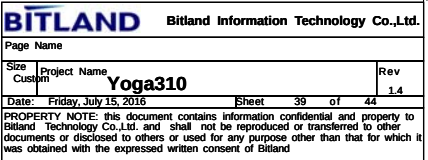




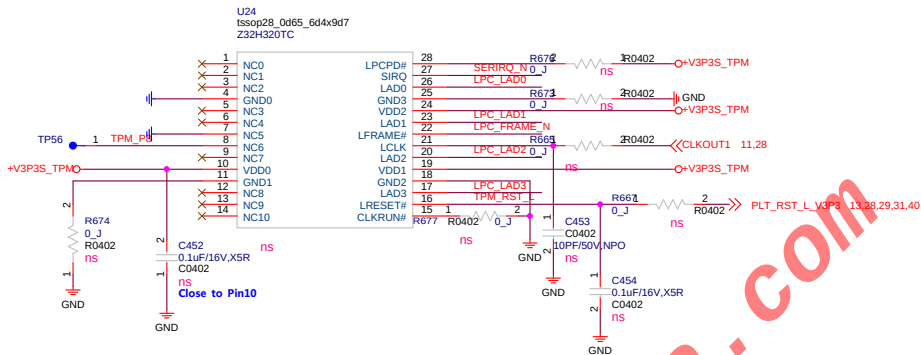
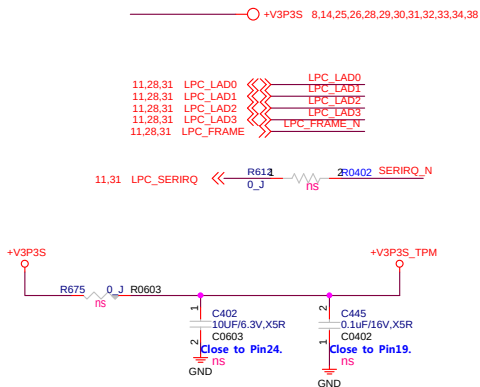


15,27,31,32,33,35 +ECVCC  
27 +VSP0A\_LDO  
34,35,37 +VDC  
27,33,37,38 +VSP0A  
11,12,13,14,25,28,29,30,33,34,37,38,40 +VSP3A

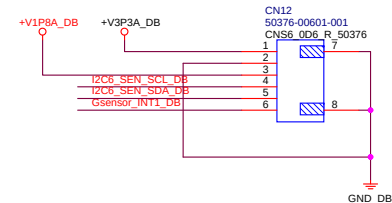
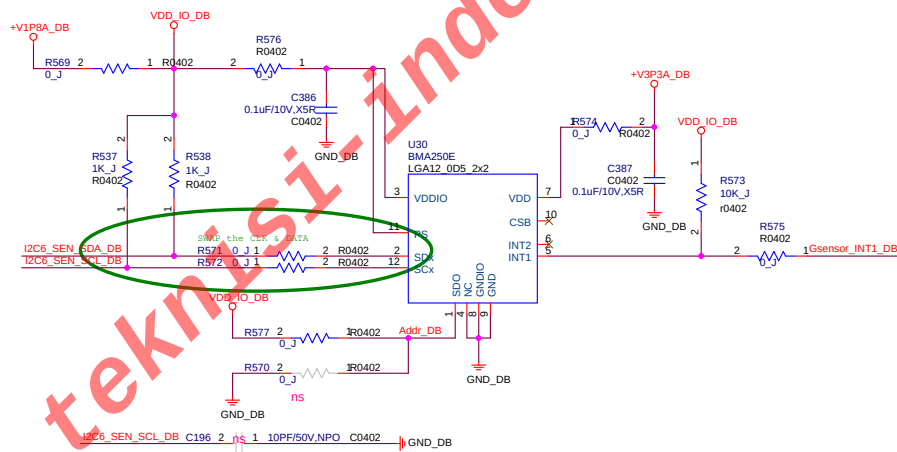




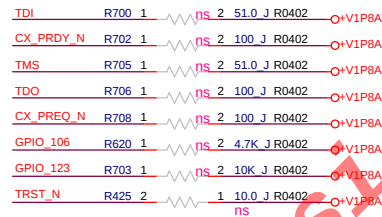
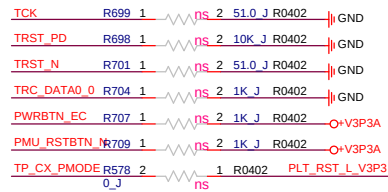
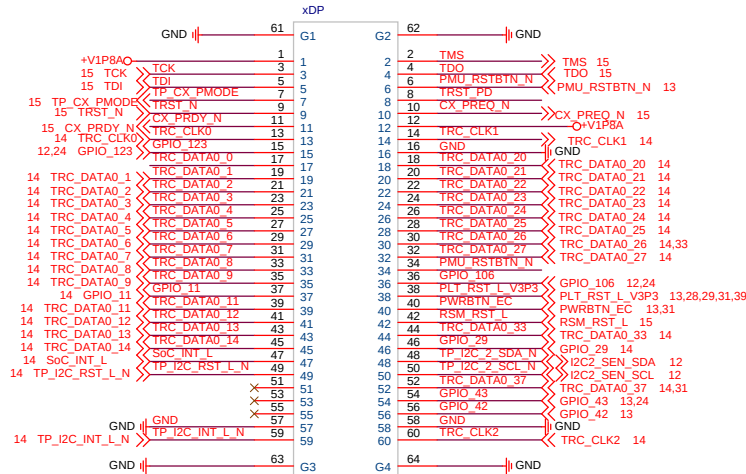


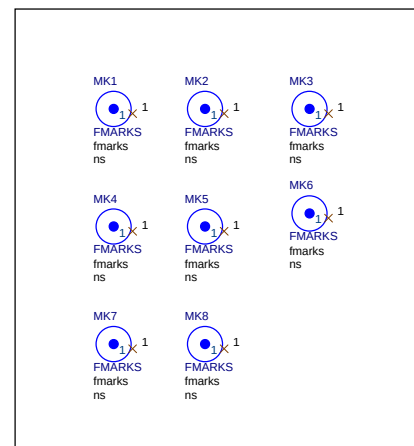
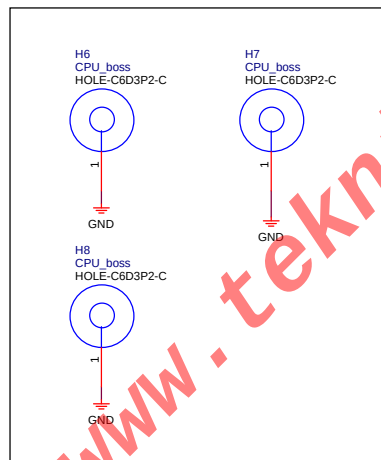
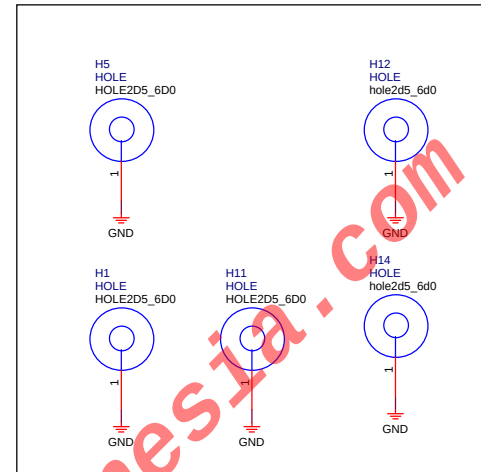
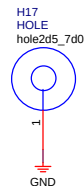
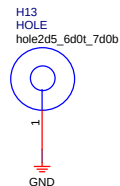
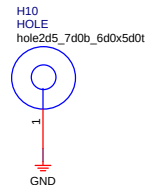
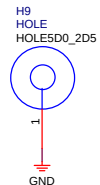
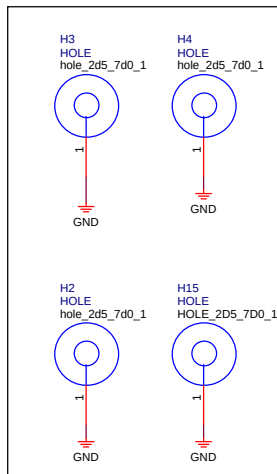


## Gsensor Card



11,12,13,14,15,18,26,28,29,30,33,34,36,37,38 +V3P3A  
8,10,11,12,13,14,15,19,24,26,27,28,29,32,33,34,37,38 +V1P8A





| Rev | Date      | Reference BM5594 V1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 | 2016/2/24 | <p>P08. Change PNL0_BKLTCTL&amp;PNL0_BKLTEN PU VOLTAGE</p> <p>P13. change R81 from 100K to 10K ,for eDP display issue</p> <p>P21. change C116 from 0.2pF to 10pF for EMC debug</p> <p>P22. change C159 from 0.2pF to 10pF for EMC debug</p> <p>P26. SD part ESD change to AZ1045-04F</p> <p>P26. change R267 to 0R for SD clock rise time fail</p> <p>P28. Del Q1&amp;Q2 , Add R377 &amp; R378 for instead</p> <p>P29. D19/D20 stuff, U17 GSG 2410 Rchanget o GSG 2450 R PF1 changet o BS4200N-C-FC For EMC solution,</p> <p>P30. change the audio codec from ALC3240 to ALC3240-VA3-CG</p> <p>P30. Change R368 from 33 ohm to 47 ohm for SDI rise skew fail</p> <p>P30. CA6,CA9,CA16,CA17,CA18,CA32 10uF change to 4.7uF</p> <p>P30. CA29 change to reserve, CA14 10uF change to 2.2uF,CA8 2.2uF change to 1uF</p> <p>P30. R91,R93 change to10 ohm, R410 change to 680 ohm for Audio HDA solution</p> <p>P31. Del R377,R378 for SMBUS0 repeating pull high</p> <p>P31. add R492 33 ohm, R390 R391 R392 0R changet o 33 ohm for S test fail</p> <p>P31. R412, R413 100R changet o 0R for SMBUS S fail</p> <p>P31. R412, R413 4 7K changet o 10K for SMBUS S fail</p> <p>P34. Change R476 from 10K to 100K,for eDP display issue</p> <p>P33. CHK10,CHK11change to stuff for EMC</p> <p>P34. Add Q40,R339,C345,C363 for camera control</p> <p>P35. Charge IC pin7, R587 changet or eservef di owvender advise</p> <p>P37. Add Q38,R72,R73 to reverse PROCHOT_L</p> <p>P18. follow 559091 change C75 from 22uF-06 to 1uF-04</p> |

| Rev | Date      | Reference BM5594 V1.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.2 | 2016/3/24 | <p>P39. swap the I2C6_SEN_SDA &amp; CLK to G-sensor</p> <p>P27. change the USB3.0 CONN to black:1011-03719</p> <p>P18. follow 559091 change C75 from 22uF-06 to 1uF-04</p> <p>P31. Add GPA4 &amp; Q41 for LID2 control</p> <p>P12.change R45/R49 from 1K to 10K for SI</p> <p>P14.change R89/R90/R95/R96 from 1K to 10K for SI</p> <p>P33.change R301,R302 to 10K,R452,R453 to 1.5K,C330,C331to ns for SI</p> <p>P30.change the CA12 to ns for HDA SI test</p> <p>P37.change the PMIC from TPS650941 to TPS650942</p> <p>P11.change USB_VBUS_SNS connect to GND to meet EDS</p> <p>P20.add 100pF-04 x30pcs for VDDQ to GND</p> <p>P14.modify the RST &amp; INTof TP for S3 issue</p> <p>P12.change pull up to V1P8S to fix the leakage from TP I2C</p> <p>P11.leave USB2_OTG_ID unconnect</p> <p>P34.change R46/R50 to 1.5K for SI</p> <p>P34.change R90/R96 to 10K for SI</p> <p>P34.change Gsensor power to S-power for I2C leakage in S3</p> <p>P34.change Touch panel power to S-power for I2C leakage in S3</p> <p>P33.change Touch pad power to S-power for I2C leakage in S3</p> <p>P31. Add GPE2 &amp; Q43 for LID1 control</p> <p>P13. add R66 10K-04 to change GPIO status to meet EDS</p> <p>P14.add R67&amp;R102 10K-04 to change GPIO status to meet EDS</p> <p>P31. add R418 0-04 for TP_LID function</p> |

| Rev | Date      | Reference BM5594 V1.2                                                                                                                                                                                                                                                         |
|-----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.3 | 2016/5/28 | <p>P08. EDP_RCOMP from 390 ohm change to 402 ohm, follow PDG</p> <p>P10. PCIE2_USB3_RCOMP from 390 ohm change to 402 ohm,follow PDG</p> <p>P16. PCIE_RCOMP from 56 ohm change to 60.4ohm ,follow PDG</p> <p>P16. M_CH0/CH1_RCOMP from 110 ohm change to 105 ohm,follow PD</p> |
| Rev | Date      | Reference BM5594 V1.3                                                                                                                                                                                                                                                         |
| 1.4 | 2016/7/11 | <p>P04. Add Power Sequence block</p> <p>P05. Add Power Sequence</p> <p>P38. Del PJ20,PJ22</p> <p>P39. Del PJ24,PJ25,PJ16,PJ21,PJ5,PJ6,PJ17,PJ15,PJ19,PJ23,PJ14,PJ18</p> <p>P43. Del EMI SHIELD</p>                                                                            |

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